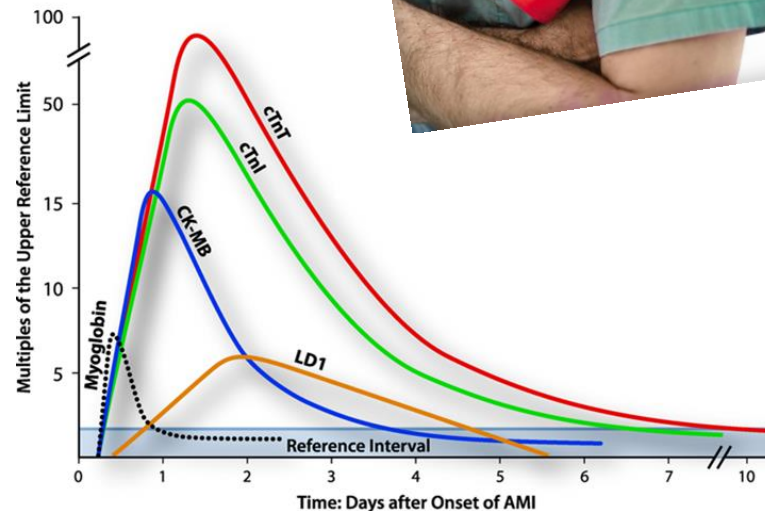


Laboratory Investigation in CVS:

Cardiac Biomarkers



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Faculty of Medicine

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Learning Objectives

วัตถุประสงค์ เมื่อสิ้นสุดการเรียนการสอน นักศึกษาสามารถ

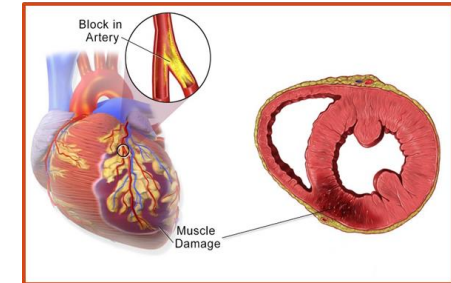
- ❑ อธิบายคุณสมบัติ ลักษณะการเปลี่ยนแปลง ข้อดี ข้อจำกัด และการใช้ประโยชน์ทางคลินิกของ cardiac biomarkers แต่ละชนิดได้อย่างถูกต้อง
- ❑ อธิบายหลักการทดสอบต่าง ๆ ทางห้องปฏิบัติการ ในการตรวจวิเคราะห์ cardiac biomarkers ได้
- ❑ เลือกใช้และแปลผลการทดสอบทางห้องปฏิบัติการในการส่งตรวจวิเคราะห์ cardiac biomarkers ได้อย่างถูกต้องและเหมาะสม



Outline

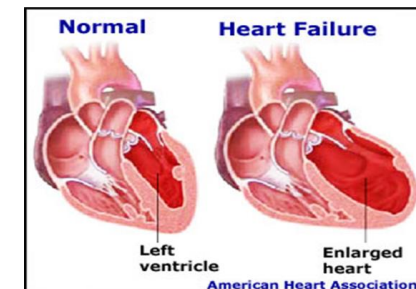
○ Biomarkers for *Myocardial Necrosis*

Acute Myocardial Infarction



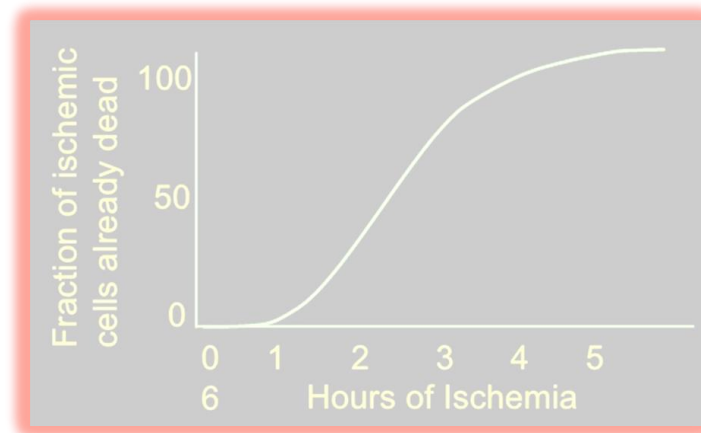
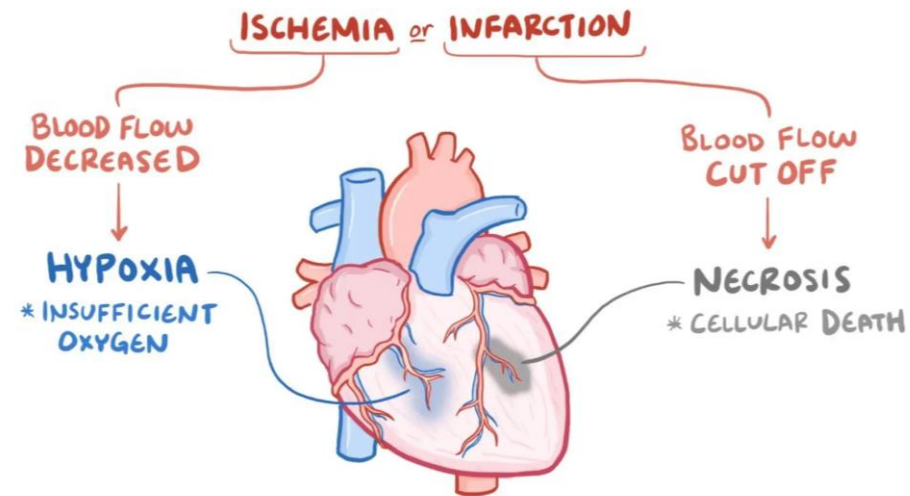
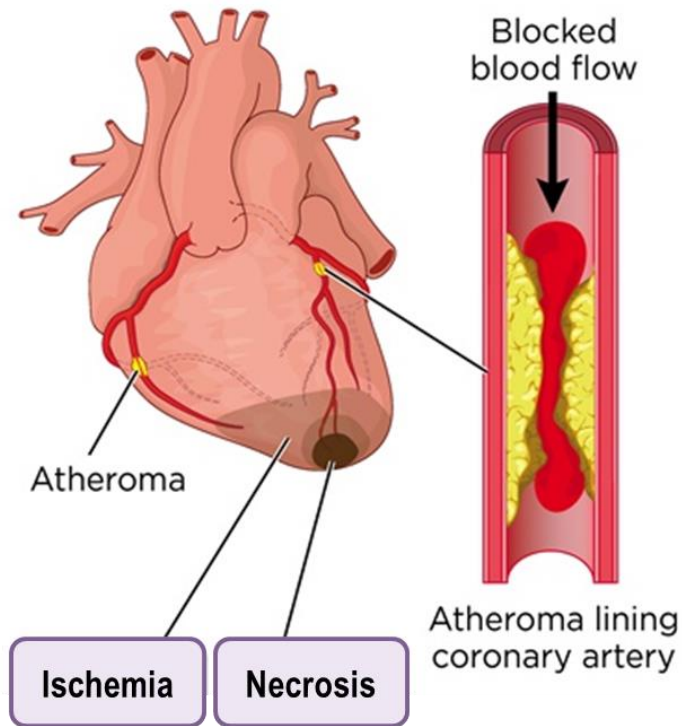
○ Biomarkers for *Myocardial Stretching* (Hemodynamic stress)

Heart Failure



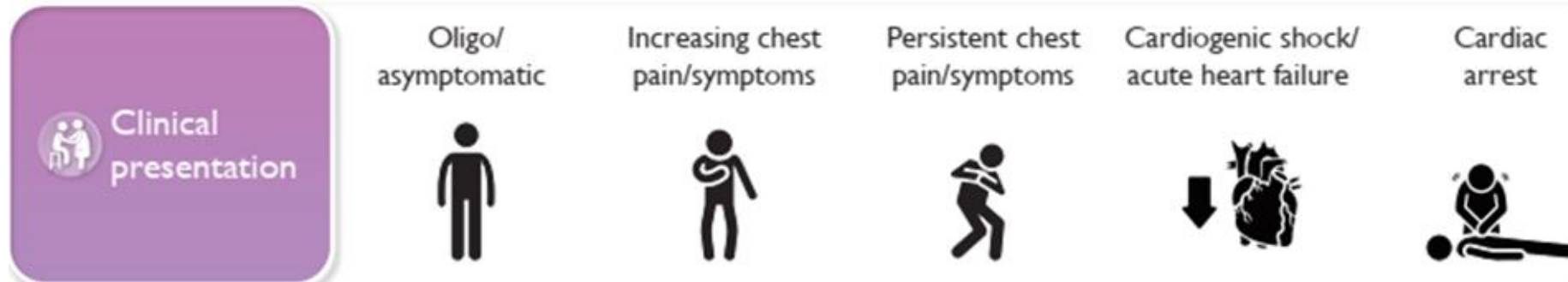
Acute Coronary Syndrome (ACS)

- A term used to describe a range of conditions associated with sudden, reduced blood flow to the heart.



Early diagnosis and therapy are important.

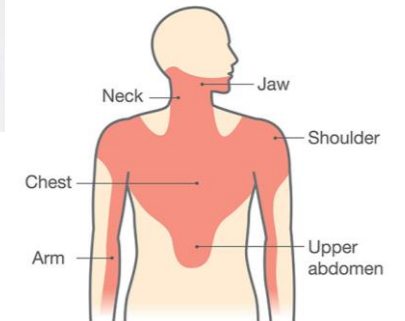
Acute Coronary Syndrome (ACS)



➤ Unstable Angina (UA)

➤ Myocardial Infarction (MI)

- Non-ST elevation Myocardial Infarction (NSTEMI)
- ST elevation Myocardial Infarction (STEMI)



Acute Coronary Syndrome (ACS)

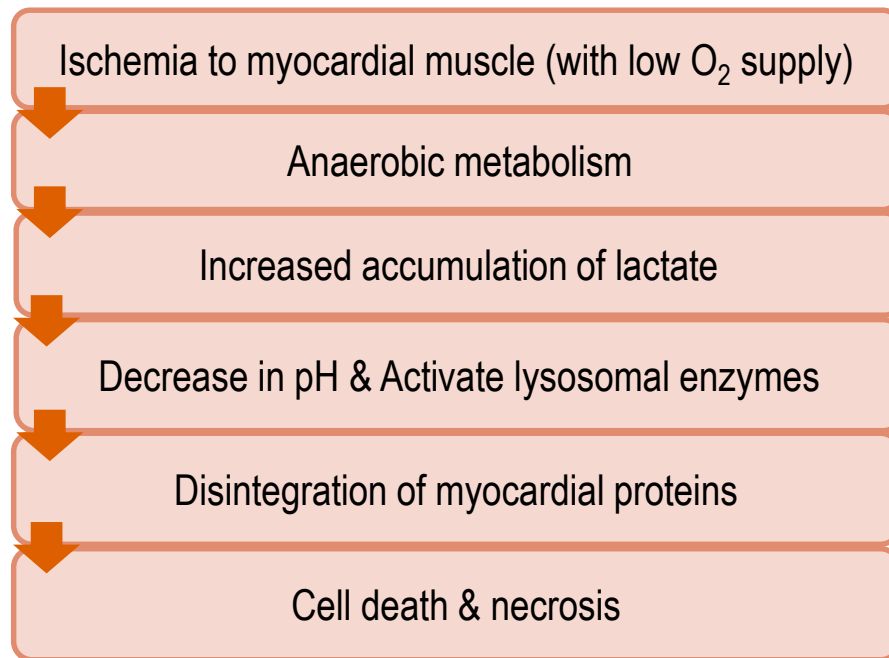
Distinguishing features of ACS

	UA	NSTEMI	STEMI
Pathophysiology	<i>Partially or transiently</i> obstructive thrombus (Incomplete occlusion)		Complete occlusion
	Ischemia <u>without</u> necrosis	Ischemia <u>with</u> necrosis	
ECG	Other ischemia change/ None		ST elevation New LBBB
Cardiac Biomarkers	Non-elevated (normal)	Elevated*	

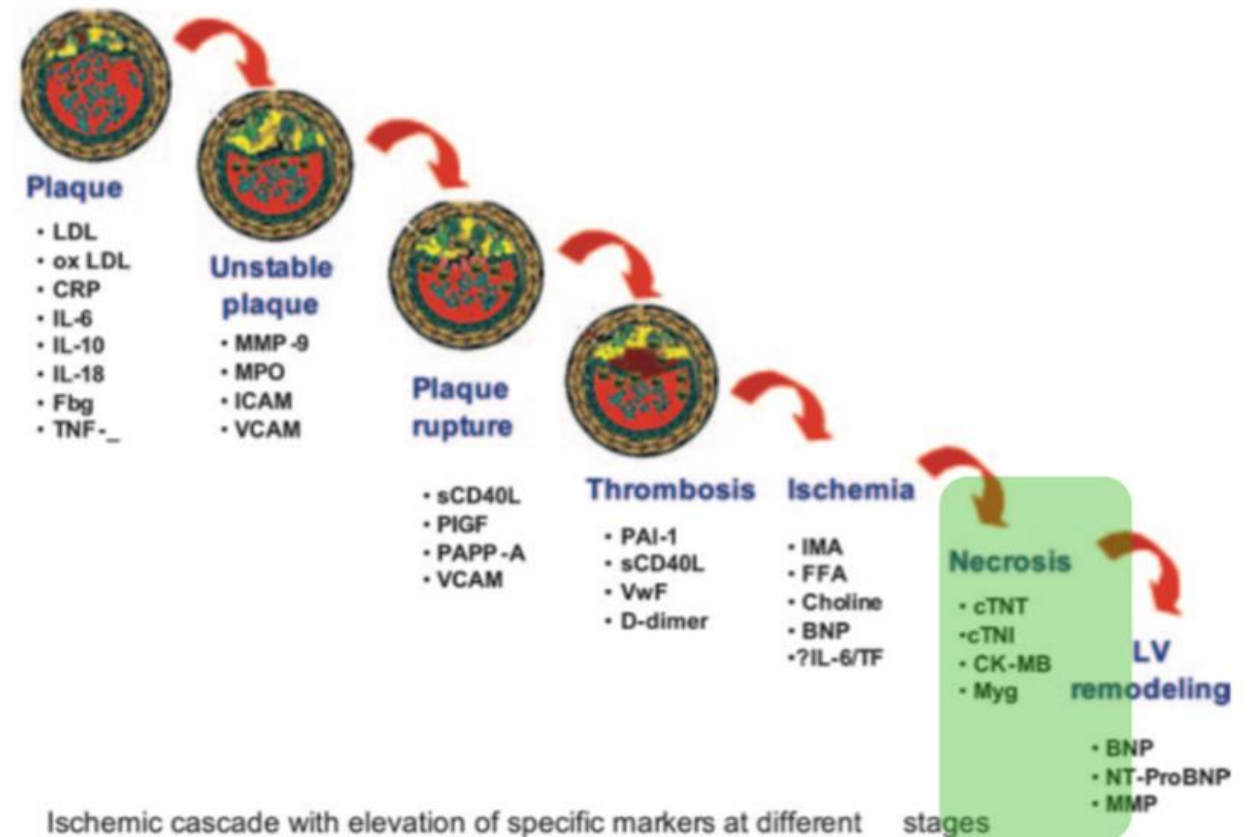


“Cardiac biomarkers” → measurable substances (proteins) released into the bloodstream from damaged cardiac myocytes

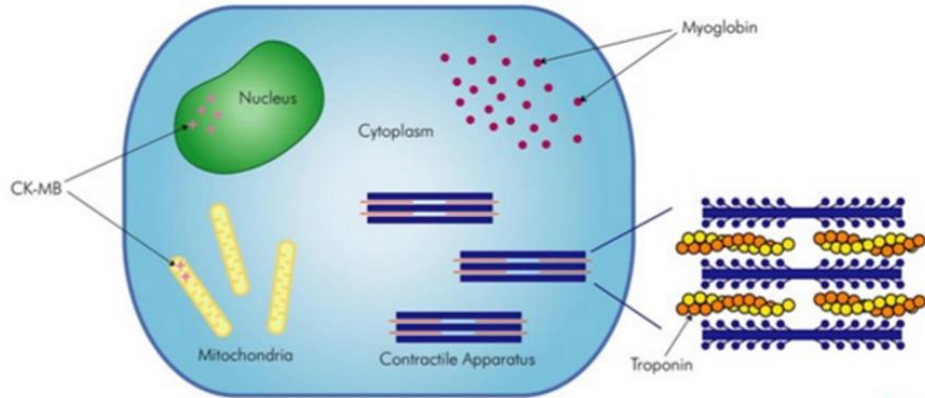
Mechanism of Biochemical Changes in AMI



Release of intracellular contents to blood



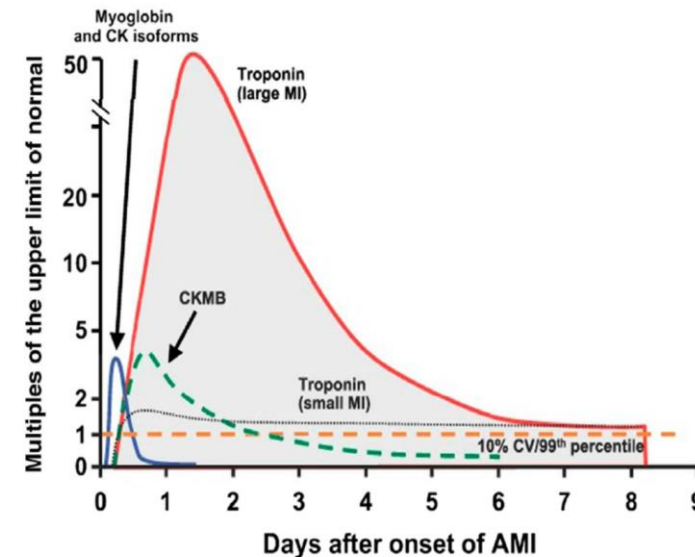
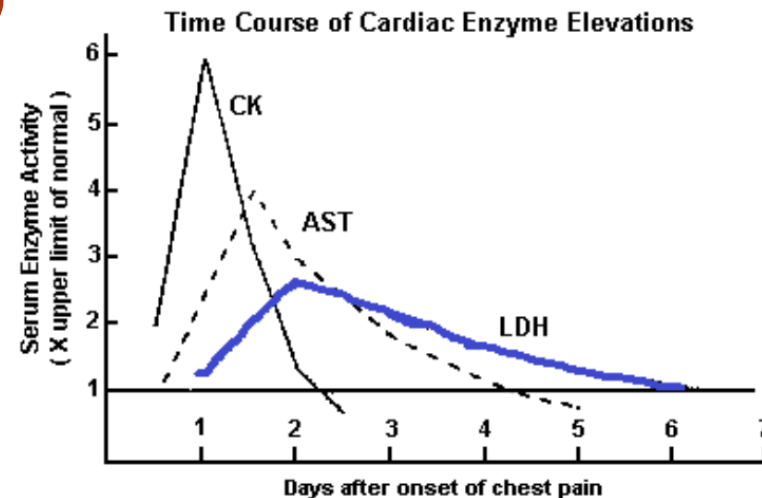
Appearance of Cardiac Biomarkers in the Bloodstream



The appearance of these biomarkers in the bloodstream and their measurable life in the blood following myocardial necrosis depends on:

- **Intracellular location:** cytosol vs the structurally bound
- **Molecular weight:** larger diffuse at a slower rate than the smaller
- **Rate of elimination from blood:** smaller rapidly eliminate as compare to larger

Rise and Fall Pattern



Cardiac Biomarkers

“Ideal” Cardiac Biomarkers

High Specificity

- ❖ High concentration in myocardium
- ❖ Absent in non-myocardial tissue
- ❖ Not detectable in blood of non-disease subject

High Sensitivity

- ❖ Released after myocardial injury:
 - Rapid release for early diagnosis
 - Long half-life in blood for late MI diagnosis
 - Shorter-time course for reinfarction diagnosis

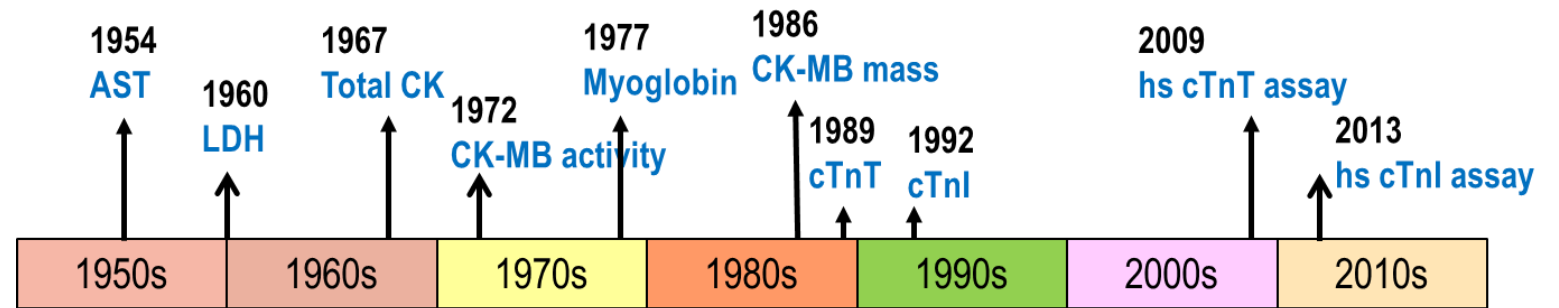
Analytical Characteristics

- ❖ Detection with the assays providing rapid, cost effective, simple, accuracy, precision

Clinical Characteristics

- ❖ Diagnosis, Treatment monitoring, Risk assessment, Prognosis of outcome

History and Timeline of Biomarkers for Myocardial Necrosis



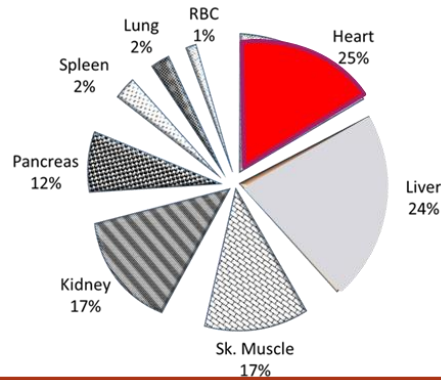
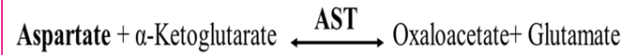
Cardiac Biomarkers in the Past

- AST (aspartate aminotransferase)
- LDH (lactate dehydrogenase)
- LDH1/LDH2 ratio
- Total CK or CPK

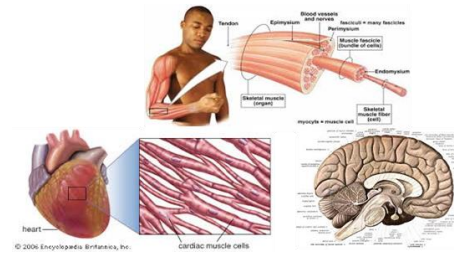
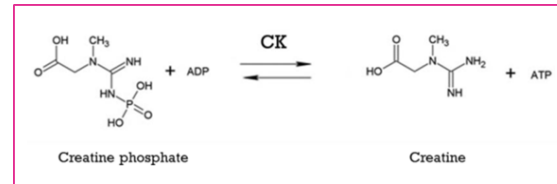
Current Cardiac Biomarkers

- CK-MB (activity and mass)
- CK-MB isoforms
- Myoglobin
- Cardiac Troponin T and I

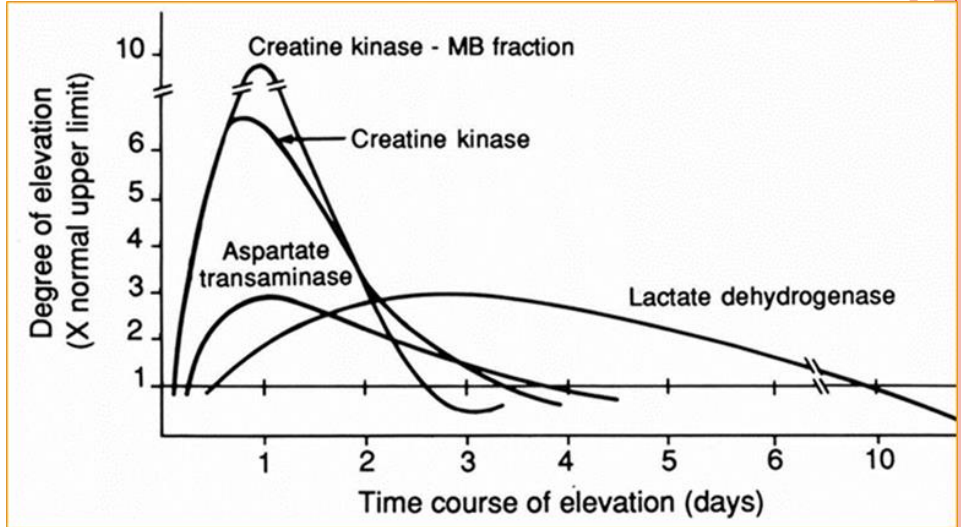
AST (Aspartate aminotransferase)



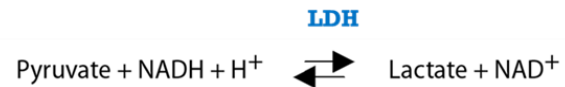
CK (Creatine kinase) CPK (Creatine phosphokinase)



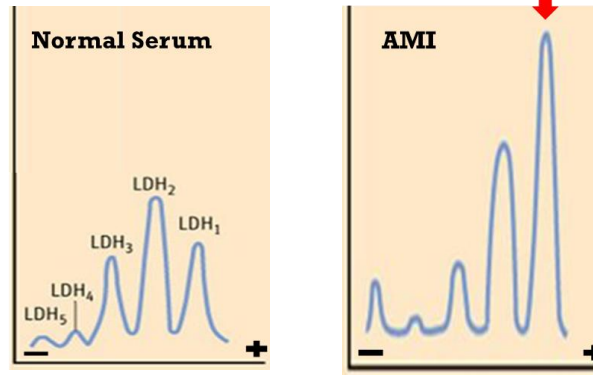
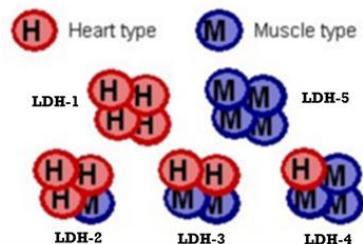
“Rise and Fall Pattern” in AMI



LDH (Lactate dehydrogenase)



➤ Heart, Liver, RBC, etc



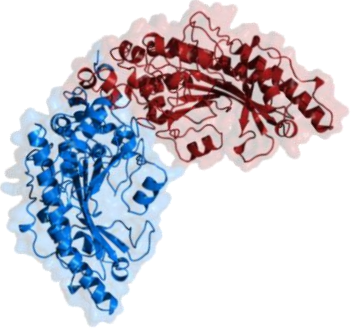
LDH-2 > LDH-1
Ratio of LDH-1/LDH-2 < 0.7

Ratio of LDH-1/LDH-2 > 1
“Flipped LDH”

Limitation: Difficult & take a long time !!!!

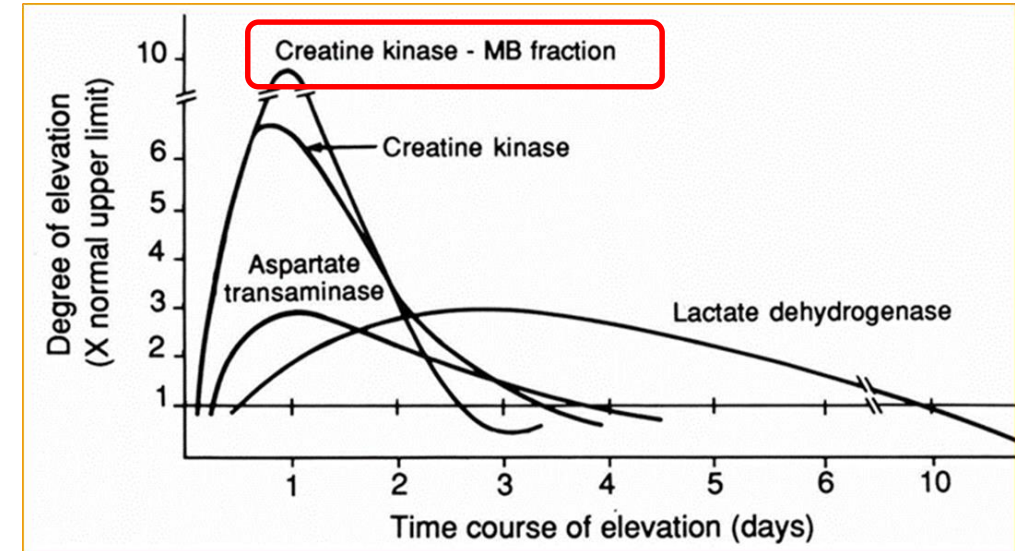
	Start to increase	Peak	Return to baseline
AST	6-12 h	24-48 h	4-6 days
LDH	8-16 h	48-72 h	6-10 days
Total CK	4-6 h	12-24 h	48-72 h

CK Isoenzyme



- CK is a dimeric protein composed of 2 subunits, **B** and **M** subunits.
- Cytosolic enzyme
- Three different isoenzymes
 - CK-BB (CK1)
 - CK-MB (CK2)
 - CK-MM (CK3)

Skeletal muscle	98% CK-MM + 2% CK-MB
Cardiac muscle	70-80% CK-MM + 20-30% CK-MB
Brain	CK-BB
Plasma/Serum	Mainly CK-MM



- Starting to **increase within 4-6 h** after onset
- Reaching the **peak in 12-24 h** after onset
- Returning to **baseline in 48-72 h** after onset

- Increased CK-MB in plasma/serum mainly comes from damaged myocardial cells.
- **More specific** than total CK

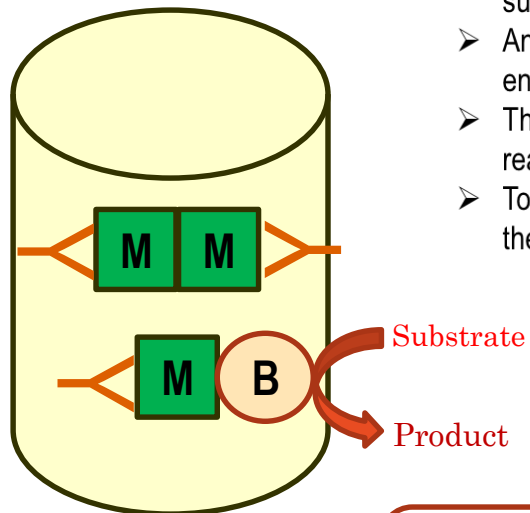


Measurement of CK-MB

CK-MB Activity

- Measure an **enzyme activity** of CK-MB
- Unit; U/L or IU/L

Immuno-inhibition method



- The sample is incubated with the antibodies against “M” subunit.
- Anti-CK-M antibodies selectively bind to and inhibit the enzymatic activity of M subunits.
- The residual CK activity, measured by using total CK reagents, is the result of the B subunits of MB alone.
- To calculate the activity of the intact CK-MB isoenzyme, the residual results must be multiplied by a factor of two.

Limitation:

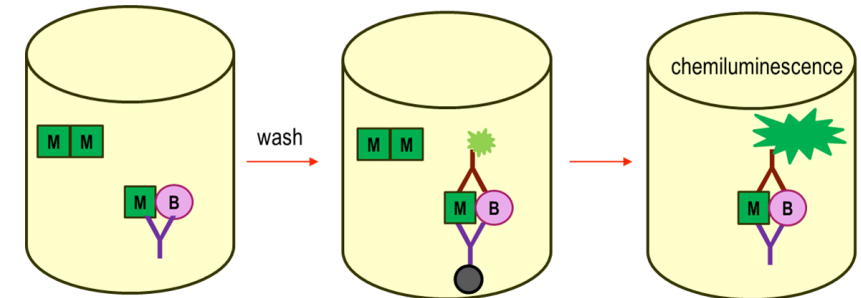
Falsely elevated results due to the presence of CK-BB and atypical CK (e.g. Macro CK)

Y anti-CK-M (specific to M subunit)

CK-MB Mass

- Measure an **amount** of CK-MB
- Unit; ng/mL or µg/L

Immunoassay



- ❖ More sensitivity than CK-MB activity
- ❖ Measured by chemiluminescence immunoassay
- can measure both *active and inactive* molecules of CK-MB

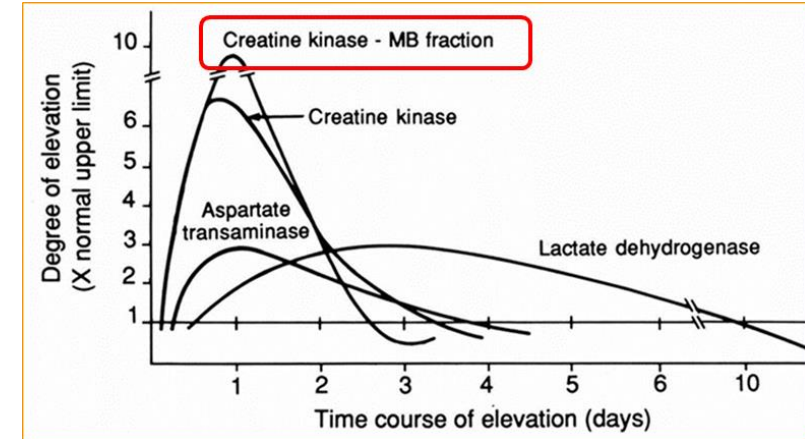
Y anti-CK-MB

Clinical Significance of CK-MB in AMI

- ❑ To calculate **the Relative Index (RI)** → *improve specificity*
- ❑ The RI is only clinically useful when both the total CK and CK-MB are increased.

$$\text{Relative index (RI)} = \frac{\text{CK-MB mass}}{\text{Total CK}} \times 100$$

- Ratio < 3% → Skeletal muscle damage
- Ratio 3-5 % → Gray zone
- Ratio > 5% → **Myocardial cell damage (suggestive MI)**

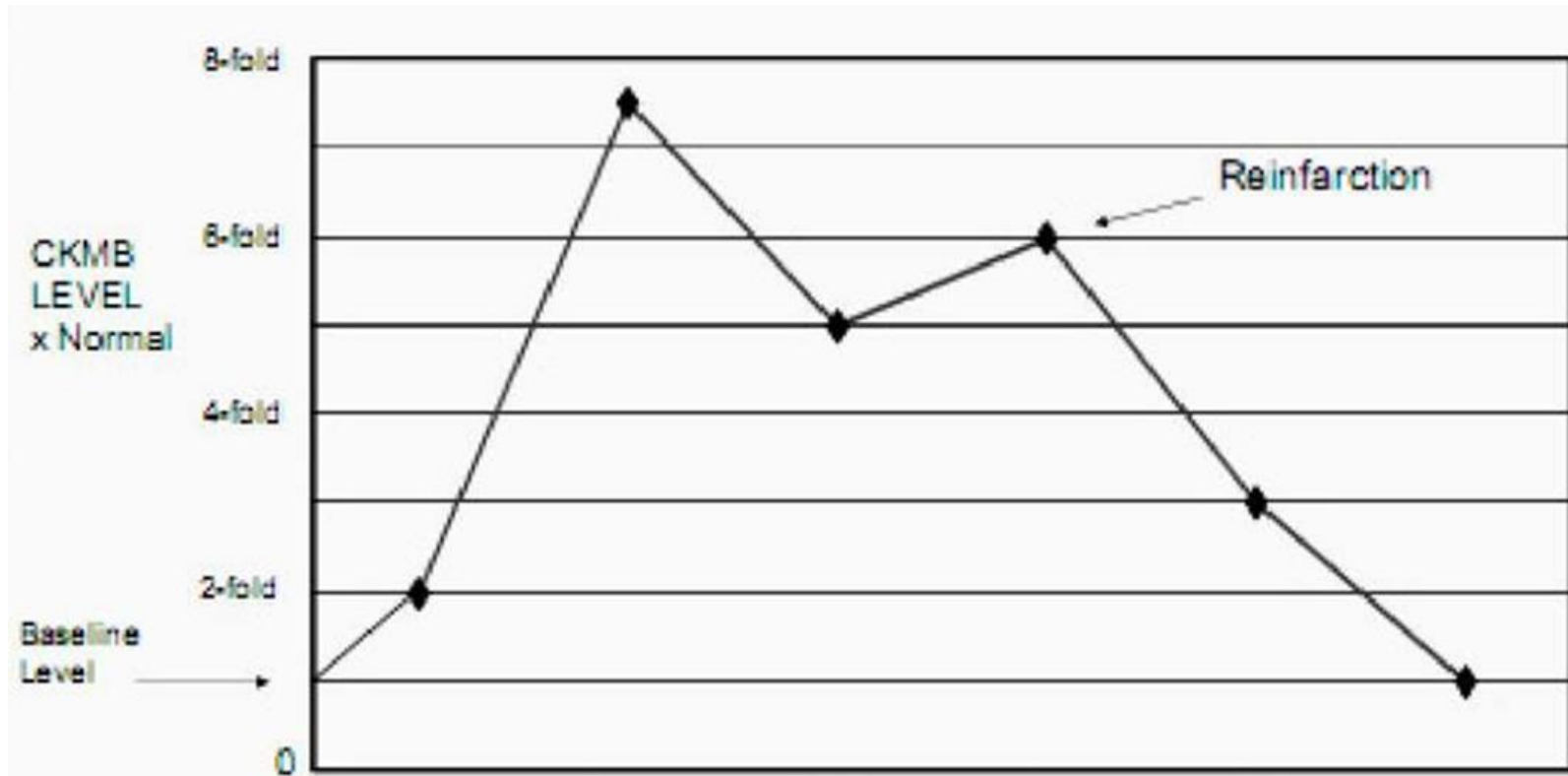


Biomarkers	Sensitivity (%)			
	2-8 h	8-24 h	24-72 h	>72 h
CK-MB	60	95	75	50

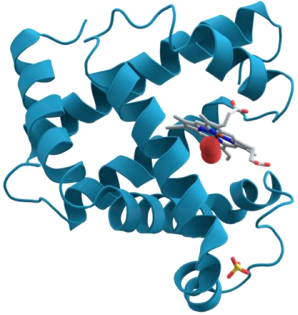
Limitation: - Not sufficient sensitivity for early diagnosis (<6 h)
- Not used for delayed admission/ late MI (>72 h)

Clinical Significance of CK-MB in AMI

Useful for Re-infarction Detection

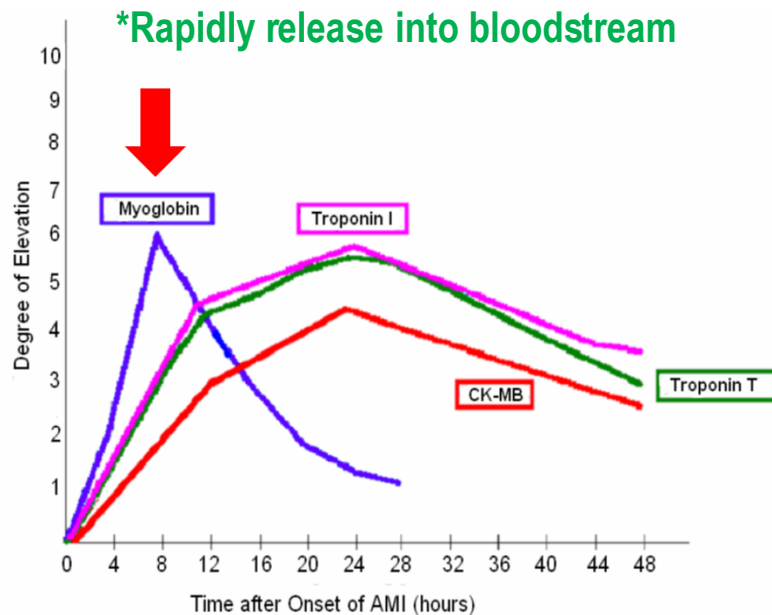


Myoglobin



- An oxygen-storing **small protein**; MW 16.7 kDa
- Present in the **cytoplasm** of all muscles

- No structural differences of Myoglobin expressed in myocardial and skeletal muscles → **Not specific** to myocardial muscle

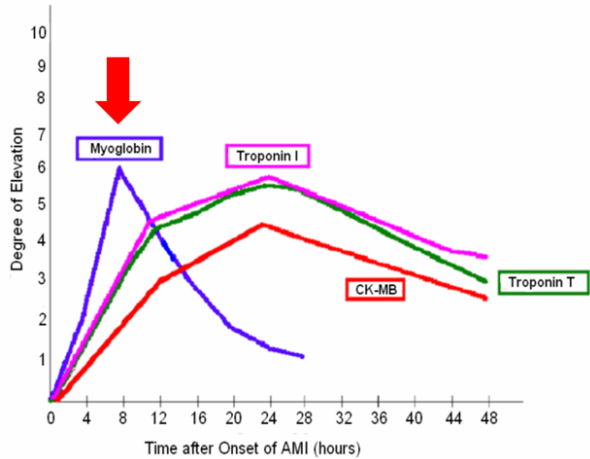


“Rise and Fall” Pattern of Myoglobin in AMI

- Starting to **increase within 1-3 h** after onset
- Reaching the **peak in 6-9 h** after onset
- Returning to **baseline level in 24-36 h** after onset

Biomarkers	Sensitivity (%)			
	2-8 h	8-24 h	24-72 h	>72 h
Myoglobin	95	75	0	0

Clinical Significance of Myoglobin in AMI



- High “negative predictive value (NPV)”

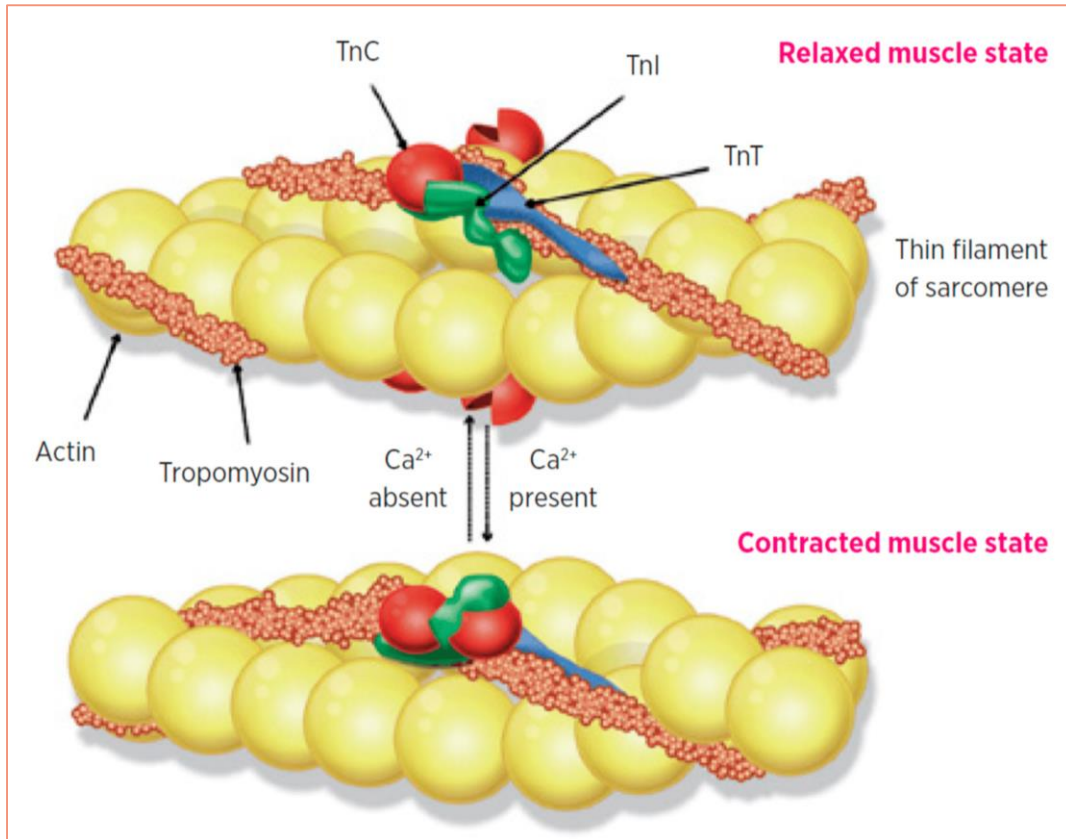
→ if myoglobin concentration remains within the reference range 8 h after onset of chest pain → potentially useful for **ruling out MI**

Limitation: Poor cardiospecificity (non-specific)

- not for confirming the diagnosis of AMI (not widely used for routine analysis)
- may use to combine measurement with CK-MB or cardiac troponin
- other causes of an increase → muscle damage, skeletal disorders and renal failure



Troponin

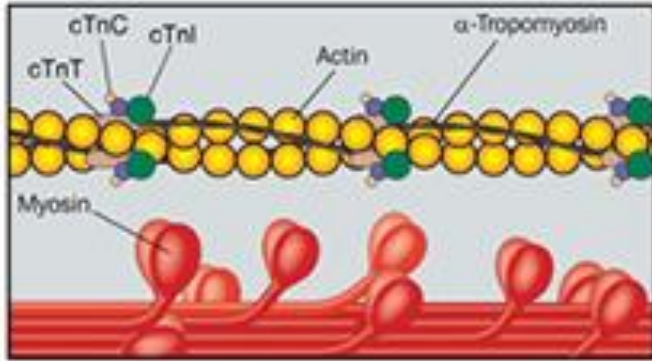


- Regulatory proteins that modulate the interaction between actin and myosin
- Troponin complex:
 - TnT**
TnI
TnC
 - Each troponin has 3 isoforms
 - Cardiac isoform
 - Slow-twitch skeletal isoform
 - Fast-twitch skeletal isoform
- sTnC & cTnC → similar amino acid sequences
- TnT & TnI of skeletal and cardiac isoforms → **dissimilarity of aa sequences**

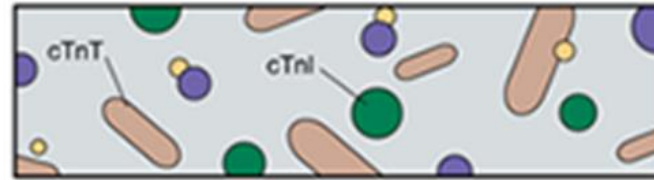
Cardiac TnT & TnI → high cardio-specificity & clinical significance

Cardiac Troponins

Myofibrillar troponins (94-97%)



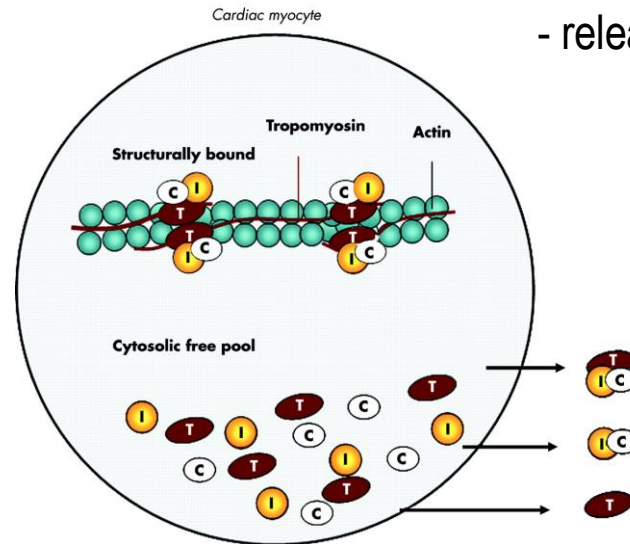
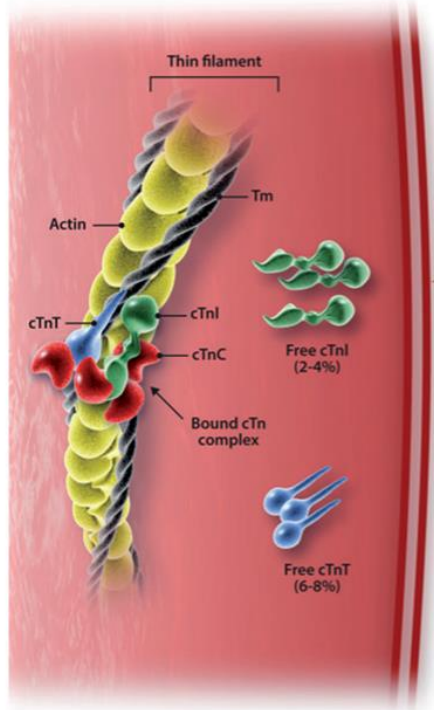
Cytosolic troponins



- Most troponins; bound to tropomyosin complex
- Small fraction in cellular cytoplasm

■ Cardiac Troponin T (cTnT): ~34 kDa

- more tissue distribution and more free cytoplasmic (6-8%) concentration
- released as complex with other cardiac troponins T-I-C and free T

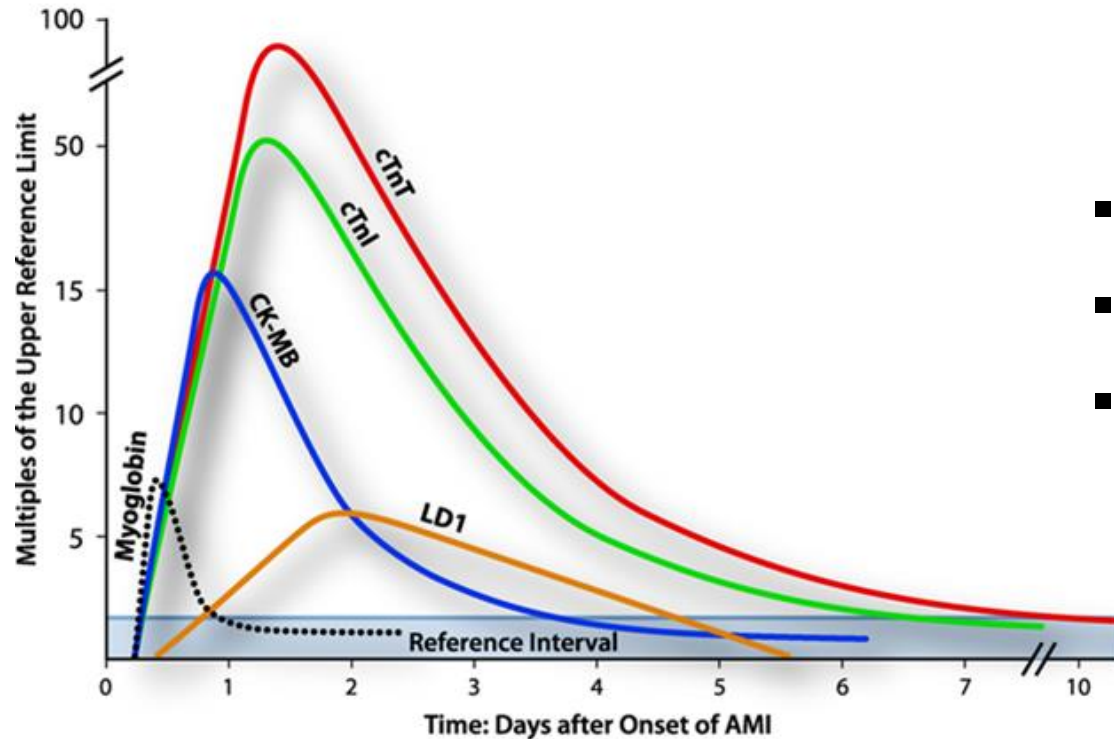


■ Cardiac Troponin I: ~24 kDa

- 2-4% free cytoplasmic concentration
- released more in the binary form; troponin I-C complex



“Rise and Fall” Pattern of cTn in AMI

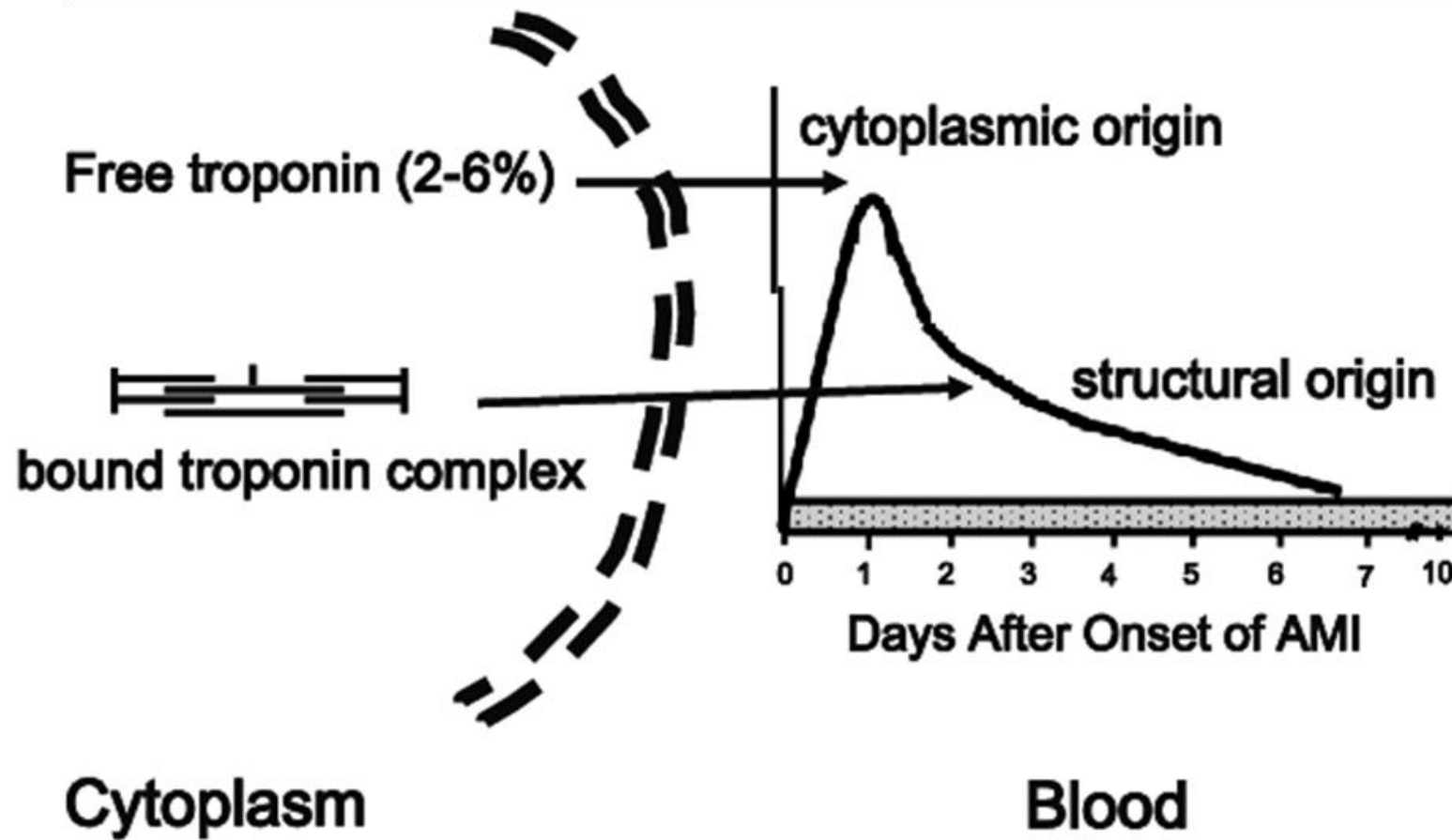


- Starting to **increase within 4-8 h** after onset
- Reaching the **peak in 12-48 h** after onset
- Returning to **baseline level in 6-10 days** after onset
[5-7 days (for cTnI) & 6-14 days (for cTnT)]

Suitable both early and late MI

Biomarkers	Sensitivity (%)			
	2-8 h	8-24 h	24-72 h	>72 h
cTn	55-65 (hs-cTn; 75-80%)	95	98	98

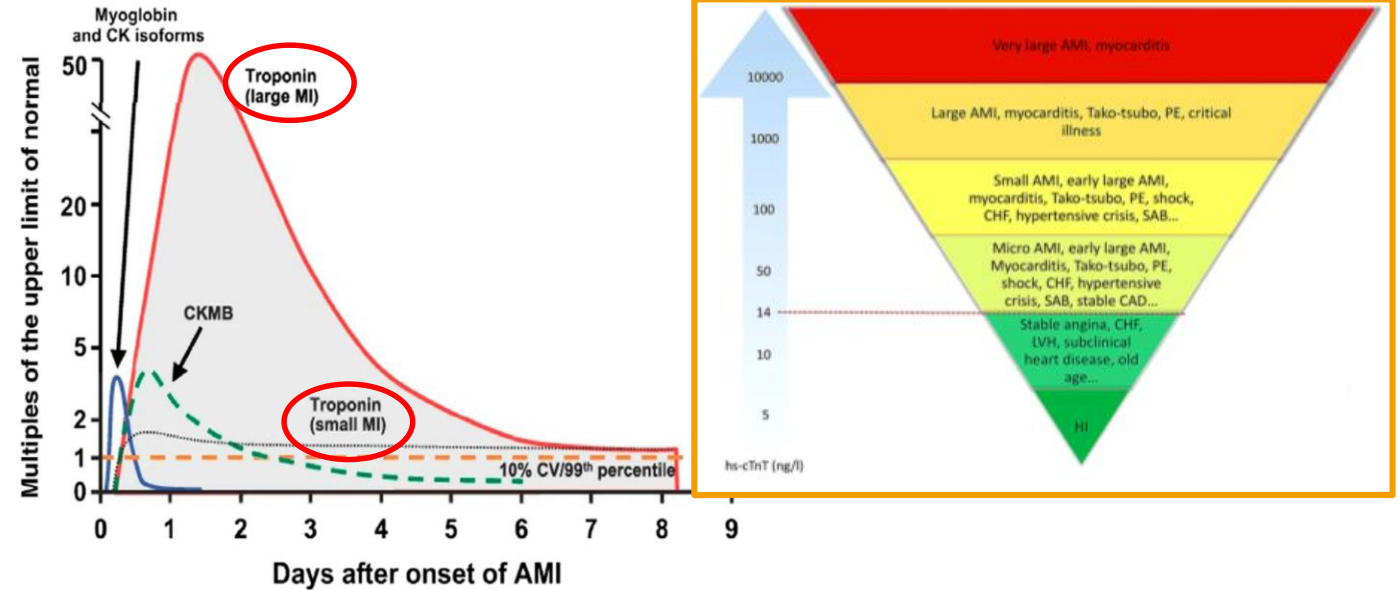
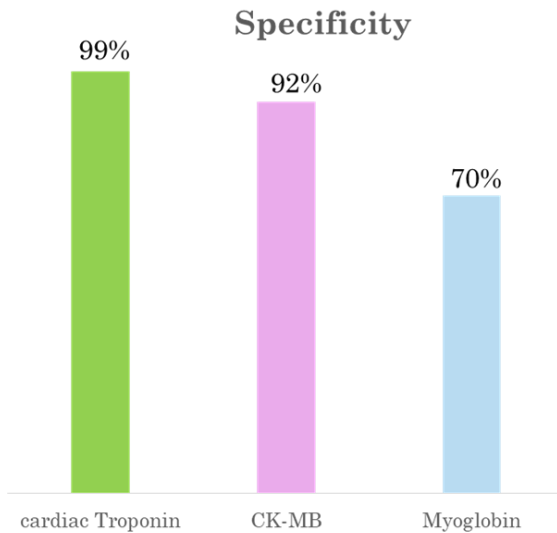
“Rise and Fall” Pattern of cTn in AMI



Clinical Significance of cTn in AMI

✓ Definitive (gold) biomarker
(high specificity & sensitivity)

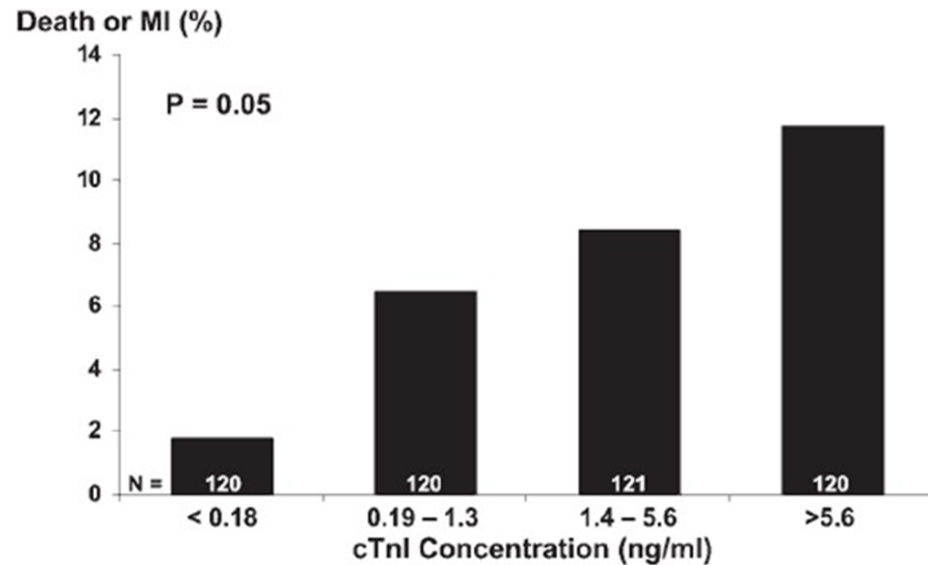
✓ Detectable the **small/minor MI** and estimation of **infarct size**



Clinical Significance of cTn in AMI

✓ Useful for **Risk Stratification and Prognosis**

- Correlation between cardiac Troponin levels and worsening outcome



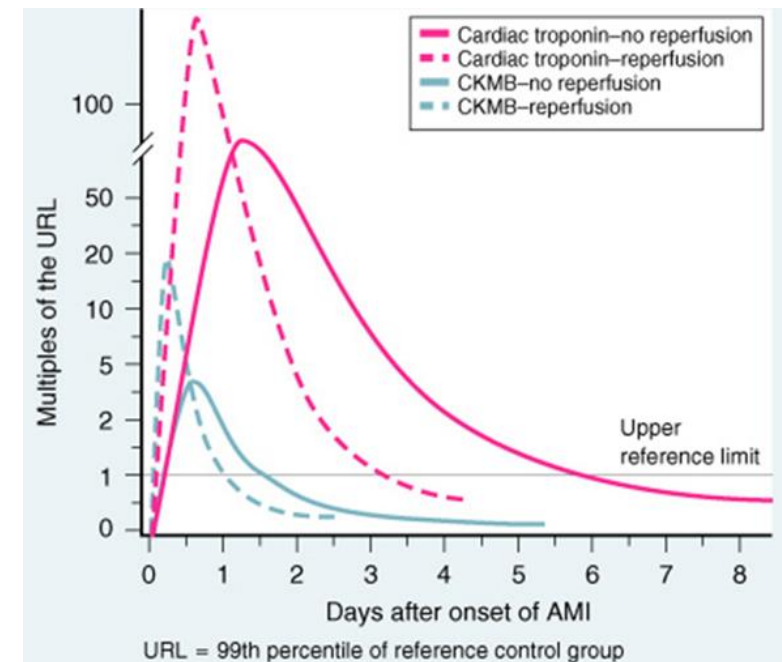
✓ Use for re-infarction detection (hs-cTn assay)

✓ Biomarker of choice for detecting AMI in patients with renal failure

✓ Monitor effectiveness of **Reperfusion** after thrombolytic therapy

- Reperfusion is the restoration of blood flow to an area after having been blocked

Reperfusion → serum concentrations of biomarkers should increase 4-7 times



POSSIBLE CAUSES OF CARDIAC TROPONIN ELEVATION

Acute myocardial infarction

Causes unrelated to acute myocardial infarction

Cardiac pathologies

- Takotsubo syndrome (Apical ballooning syndrome)
- Cardiac inflammation (endocarditis, myocarditis, pericarditis)
- Hereditary cardiomyopathies
- Heart failure
- Cardiac infiltrative disorders (amyloidosis, sarcoidosis, etc.)
- Arrhythmias
- Cardiac Contusion
- Cardiac Ablation
- Cardiotoxic drugs (anthracyclines, anthraquinones, antimetabolites, methamphetamine etc.)

Non cardiac and systemic pathologies

- Pulmonary embolism
- Chronic obstructive pulmonary disease
- Hypertension
- Acute aortic dissection
- Chronic kidney diseases
- Systemic inflammatory reaction (sepsis)
- Diabetes mellitus
- Oncological diseases
- Acute stroke (ischemic stroke, hemorrhagic stroke)
- Severe bleeding and/or anemia
- COVID-19

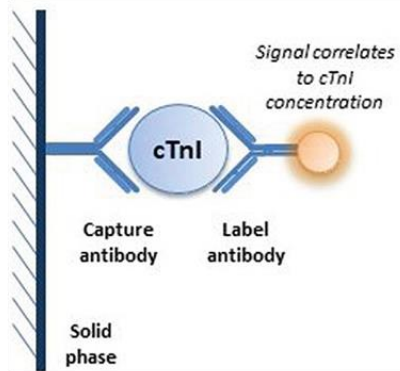
False positive causes

- Fibrin clots
- Heterophile antibodies
- Alkaline phosphatase
- Rheumatoid factor
- Cross-reactions of diagnostic (anti-cTn) antibodies with troponin molecules released from skeletal muscle

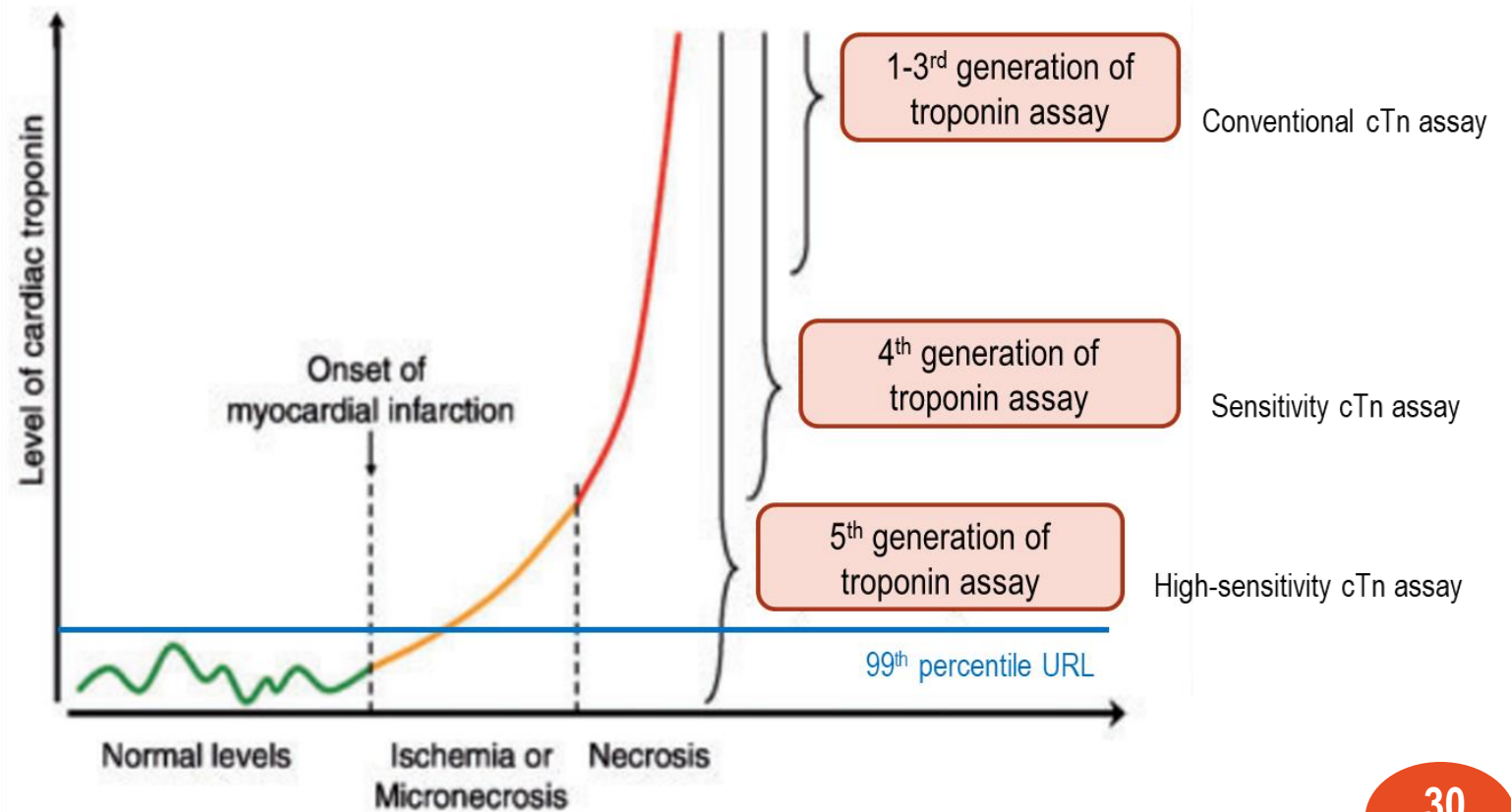
cTn Assays



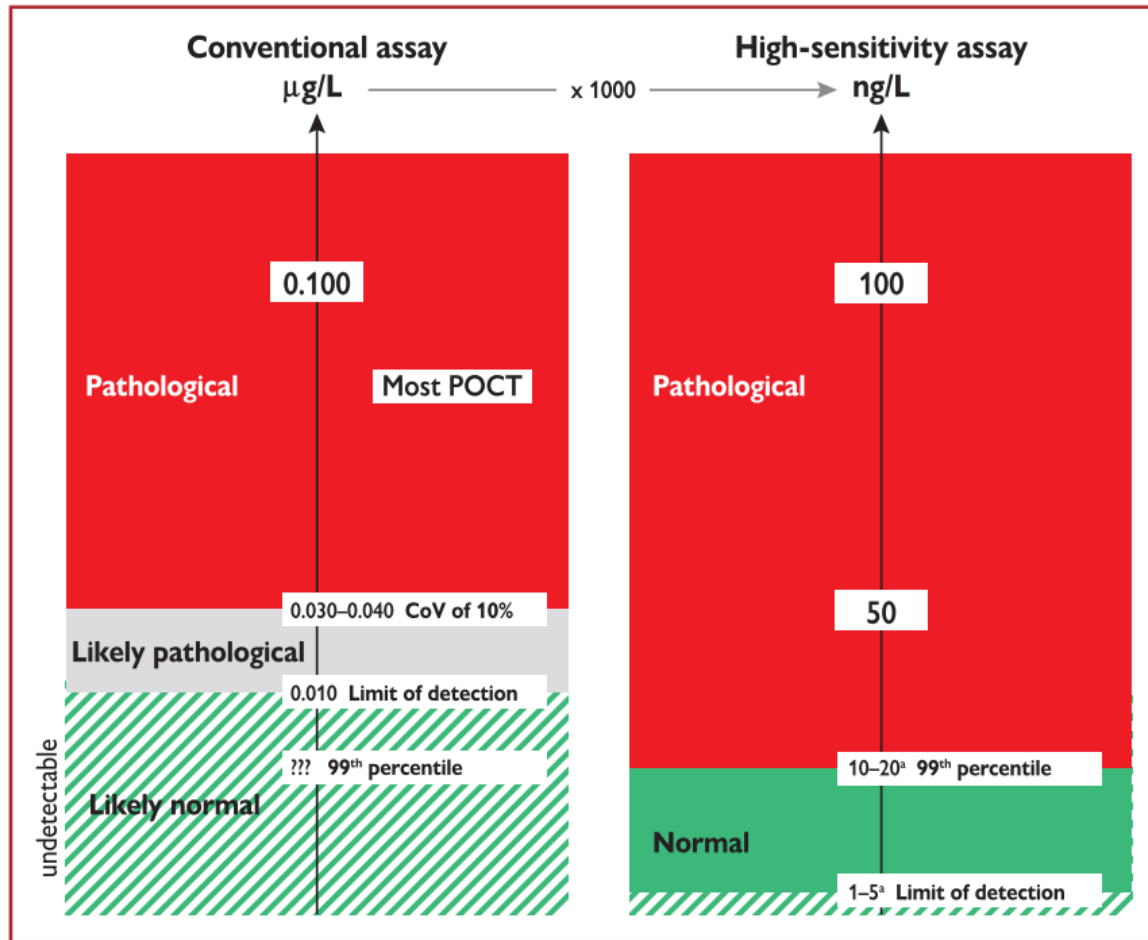
Immunoassay



Improving Analytical Sensitivity of cTn Assay



Improving Analytical Sensitivity of cTn Assay



High-sensitivity cardiac troponin (hs-cTn)

- High analytical sensitivity (detection of low concentration)
- Results are expressed in the range of single digits of ng/L
- Increase “sensitivity” for early detection
- Improving diagnostic accuracy (improving NPV)

Although hs-cTn assays detect cTn release earlier than the conventional cTn assays, ***time-to-peak blood levels and fall pattern are still the same.***

Table 3. FDA-Cleared High-Sensitivity Cardiac Troponin Assay Thresholds

Assay	LoD, ng/L	LoQ, ng/L	Overall 99th percentile, ng/L	Sex-specific 99thpercentiles, F/M, ng/L
Abbott ARCHITECT hs-cTnI	1.7	2.3	28	17/35
Beckman Coulter Access 2 hs-cTnI (plasma)	1.0–2.0	0.9–2.3	17.5	11.6/19.8
Beckman Coulter Access 2 hs-cTnI (serum)	1.0–2.0	0.9–2.3	18.2	11.8/19.7
Beckman Coulter Dxl Access hs-cTnI (plasma)	1.5–2.3	1.2–2.3	17.9	14.9/19.8
Beckman Coulter Dxl Access hs-cTnI (serum)	1.5–2.3	1.2–2.3	18.1	13.6/19.8
Roche cobas e601, e602, E170/TnT Gen 5 STAT	3; 5 for e411	6	19	14/22
Siemens ATELLICA high-sensitivity TnI (TNIH)	1.6	2.50	45.4	38.6/53.5
Siemens ADVIA Centaur XP/XPT/CP high-sensitivity TnI (TNIH)	1.6	2.50	46.5	39.6/58.0
Siemens Dimension VISTA high-sensitivity TnI (TNIH)	2.0	3.0	58.9	53.7/78.5
Siemens Dimension ExL high-sensitivity TnI (TNIH)	2.7	4.0	60.4	51.4/76.2

Limit of detection (LoD), limit of quantitation (LoQ), and 99th percentile upper reference limits according to the insert package information. Source: reference 10. Beckman-Coulter has chosen to report their LoD as equal to the LoQ for ease of reporting. FDA indicates Food and Drug Administration; and hs-cTnI, high-sensitivity cardiac troponin I.

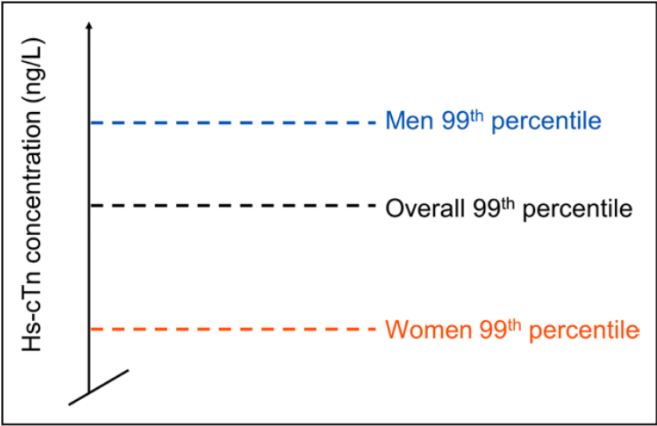
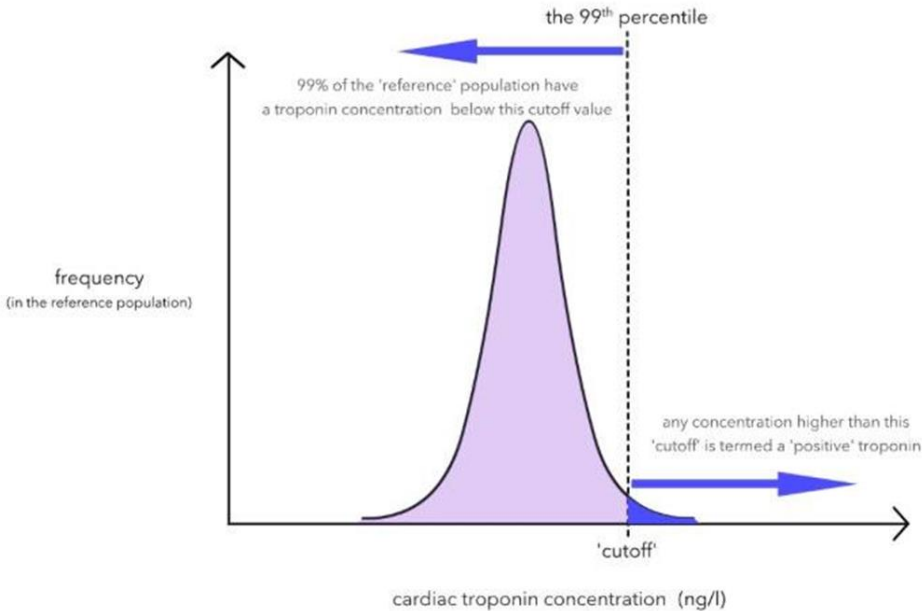
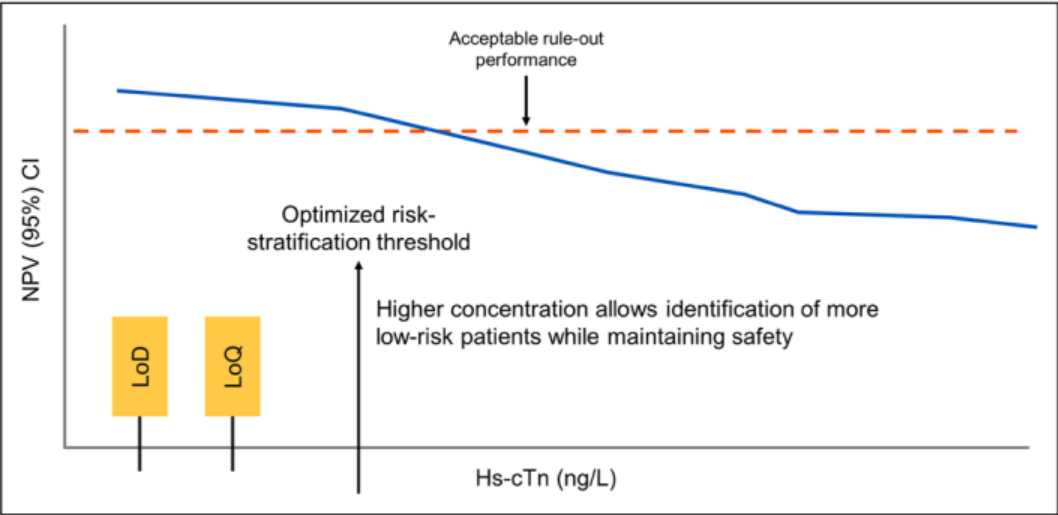
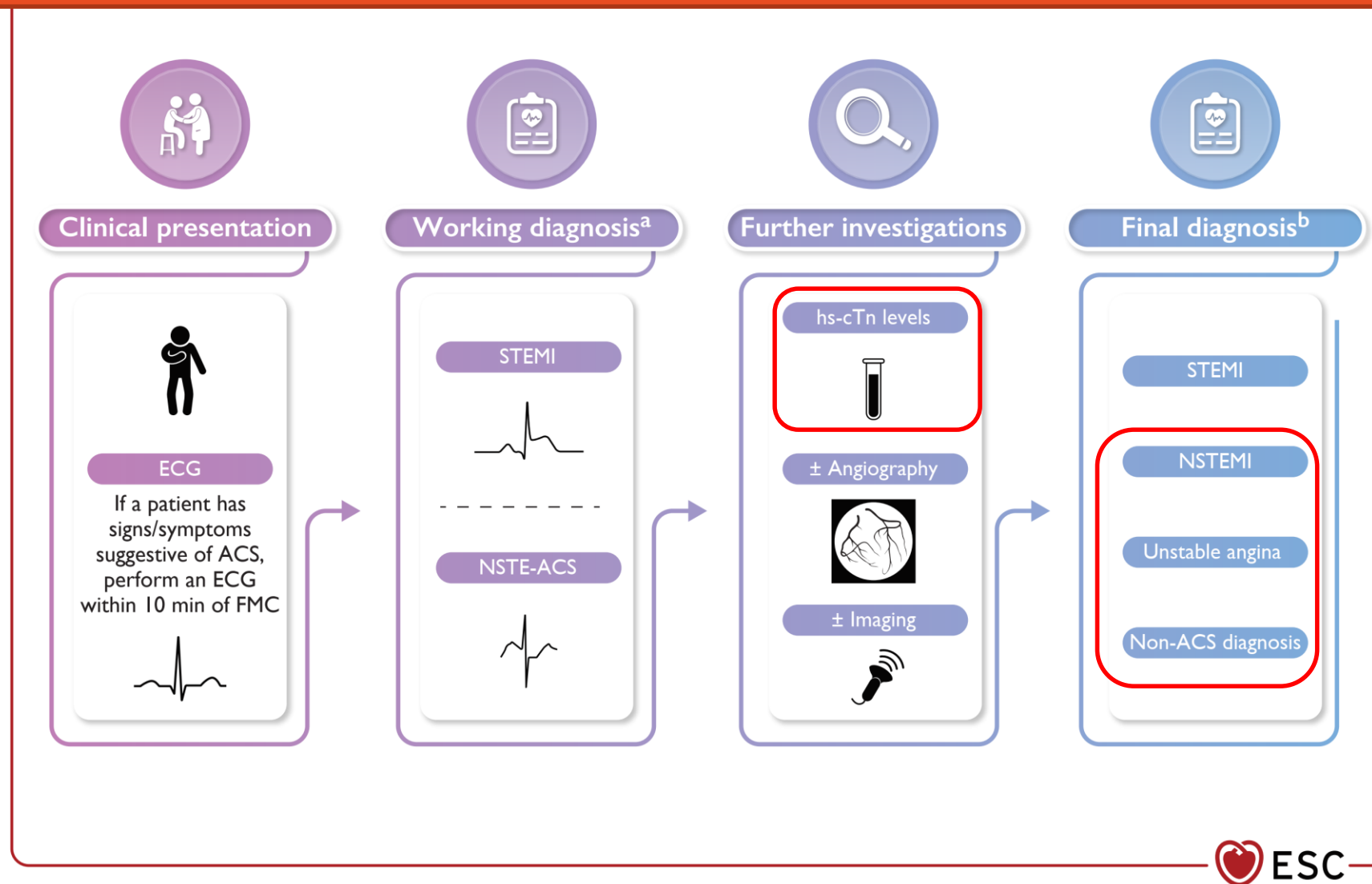


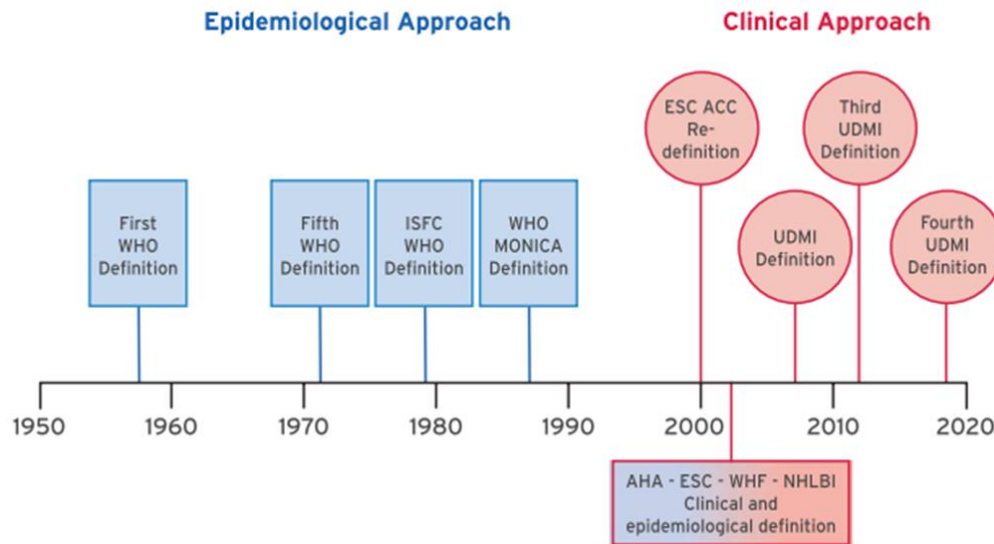
Figure 2. Sex-specific 99th percentile upper-reference limits for high-sensitivity cardiac troponin assays.



2023 ESC Guidelines for the Management of Acute Coronary Syndromes



History of Documents on the Definition of Myocardial Infarction



Fourth Universal Definition of Myocardial Infarction

Universal definitions of myocardial injury and myocardial infarction

Criteria for myocardial injury

The term myocardial injury should be used when there is evidence of elevated cardiac troponin values (cTn) with at least 1 value above the 99th percentile upper reference limit (URL). The myocardial injury is considered acute if there is a rise and/or fall of cTn values.

Criteria for acute myocardial infarction (types 1, 2 and 3 MI)

The term acute myocardial infarction should be used when there is acute myocardial injury with clinical evidence of acute myocardial ischemia and with detection of a rise and/or fall of cTn values with at least 1 value above the 99th percentile URL and at least 1 of the following:

- Symptoms of myocardial ischemia;
- New ischemic ECG changes;
- Development of pathological Q waves;
- Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality in a pattern consistent with an ischemic etiology;
- Identification of a coronary thrombus by angiography or autopsy (not for types 2 or 3 MIs).

Postmortem demonstration of acute atherothrombosis in the artery supplying the infarcted myocardium meets criteria for *type 1 MI*. Evidence of an imbalance between myocardial oxygen supply and demand unrelated to acute atherothrombosis meets criteria for *type 2 MI*. Cardiac death in patients with symptoms suggestive of myocardial ischemia and presumed new ischemic ECG changes before cTn values become available or abnormal meets criteria for *type 3 MI*.

Criteria for coronary procedure-related myocardial infarction (types 4 and 5 MI)

Percutaneous coronary intervention (PCI)-related MI is termed *type 4a MI*.

Coronary artery bypass grafting (CABG)-related MI is termed *type 5 MI*.

Coronary procedure-related MI ≤48 hours after the index procedure is arbitrarily defined by an elevation of cTn values >5 times for *type 4a MI* and >10 times for *type 5 MI* of the 99th percentile URL in patients with normal baseline values. Patients with elevated preprocedural cTn values, in whom the preprocedural cTn level are stable (≤20% variation) or falling, must meet the criteria for a >5 or >10 fold increase and manifest a change from the baseline value of >20%. In addition with at least 1 of the following:

- New ischemic ECG changes (this criterion is related to *type 4a MI* only);
- Development of new pathological Q waves;
- Imaging evidence of loss of viable myocardium that is presumed to be new and in a pattern consistent with an ischemic etiology;
- Angiographic findings consistent with a procedural flow-limiting complication such as coronary dissection, occlusion of a major epicardial artery or graft, side-branch occlusion-thrombus, disruption of collateral flow or distal embolization.

Isolated development of new pathological Q waves meets the *type 4a MI* or *type 5 MI* criteria with either revascularization procedure if cTn values are elevated and rising but less than the prespecified thresholds for PCI and CABG.

Other types of 4 MI include *type 4b MI* stent thrombosis and *type 4c MI* restenosis that both meet *type 1 MI* criteria.

Postmortem demonstration of a procedure-related thrombus meets the *type 4a MI* criteria or *type 4b MI* criteria if associated with a stent.

Criteria for prior or silent/unrecognized myocardial infarction

Any 1 of the following criteria meets the diagnosis for prior or silent/unrecognized MI:

- Abnormal Q waves with or without symptoms in the absence of nonischemic causes.
- Imaging evidence of loss of viable myocardium in a pattern consistent with ischemic etiology.
- Patho-anatomical findings of a prior MI.

CABG indicates coronary artery bypass grafting; cTn, cardiac troponin; ECG, electrocardiogram; MI, myocardial infarction; PCI, percutaneous coronary intervention; URL, upper reference limit.

Universal definitions of myocardial injury and myocardial infarction

Criteria for myocardial injury

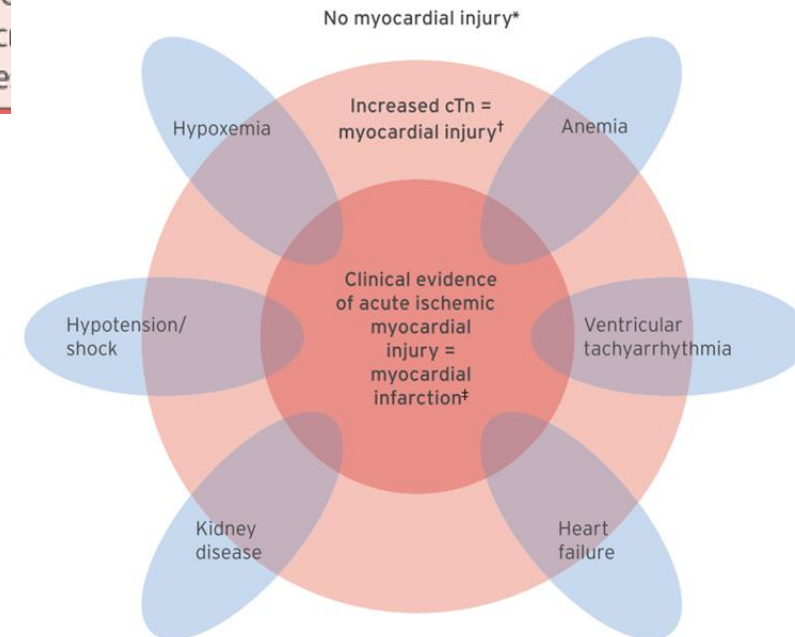
The term myocardial injury should be used when there is evidence of elevated cardiac troponin values (cTn) with at least 1 value above the 99th percentile upper reference limit (URL). The myocardial injury is considered acute if there is a rise and/or fall of cTn values.

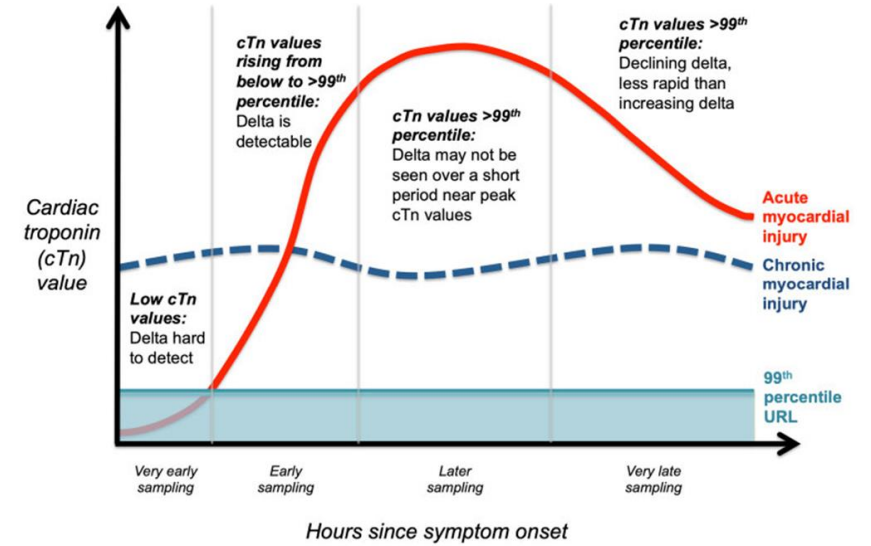
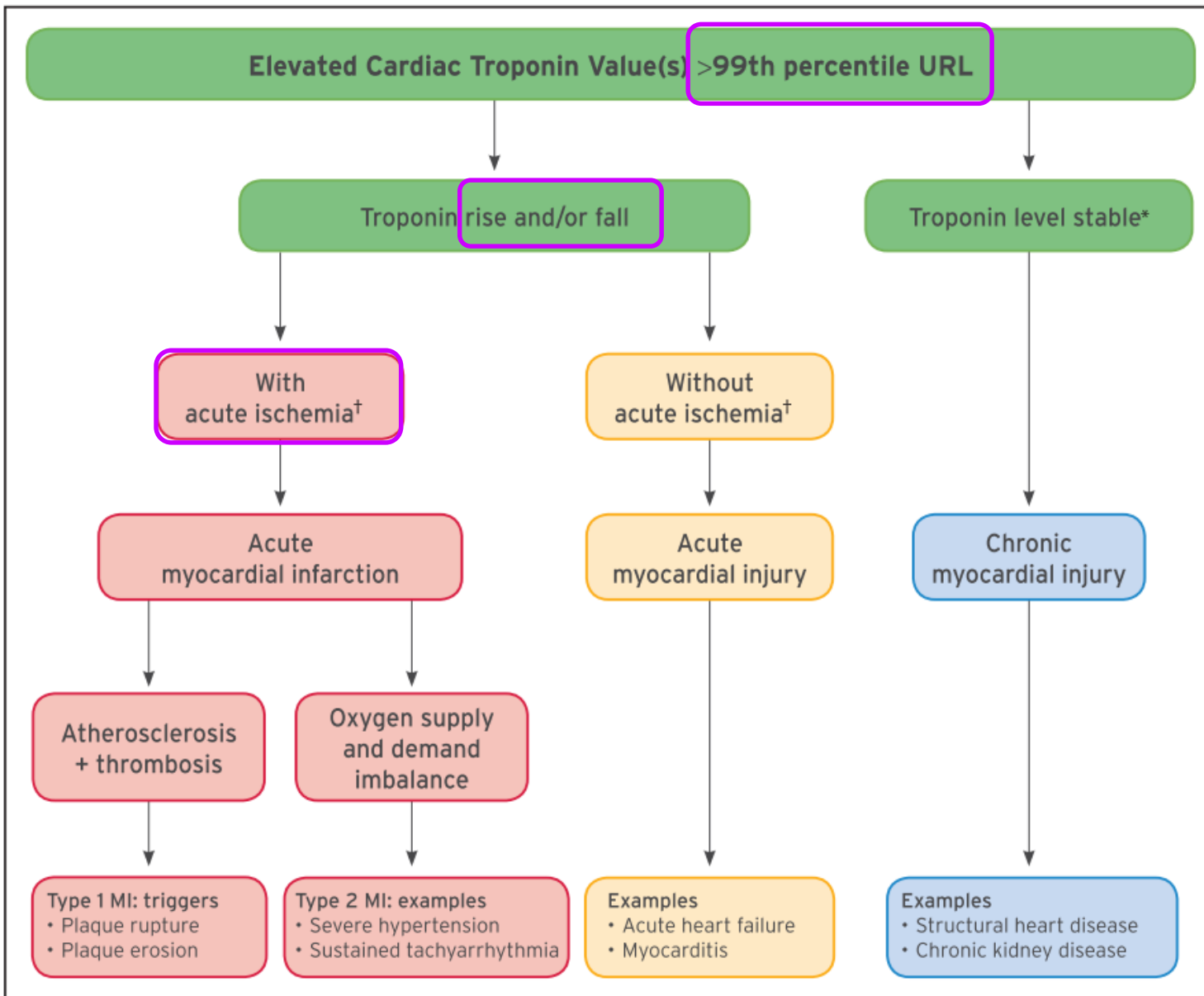
Criteria for acute myocardial infarction (types 1, 2 and 3 MI)

The term acute myocardial infarction should be used when there is acute myocardial injury with clinical evidence of acute myocardial ischemia and with detection of a rise and/or fall of cTn values with at least 1 value above the 99th percentile URL and at least 1 of the following:

- Symptoms of myocardial ischemia;
- New ischemic ECG changes;
- Development of pathological Q waves;
- Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality in a pattern consistent with an ischemic etiology;
- Identification of a coronary thrombus by angiography or autopsy (not for types 2 or 3 MIs).

Postmortem demonstration of acute atherothrombosis in the artery supplying the infarcted myocardium meets criteria for *type 1 MI*. Evidence of an imbalance between myocardial oxygen supply and demand unrelated to acute atherothrombosis meets criteria for *type 2 MI*. Evidence of an imbalance suggestive of myocardial ischemia and presumed new ischemic ECG changes before cTn value



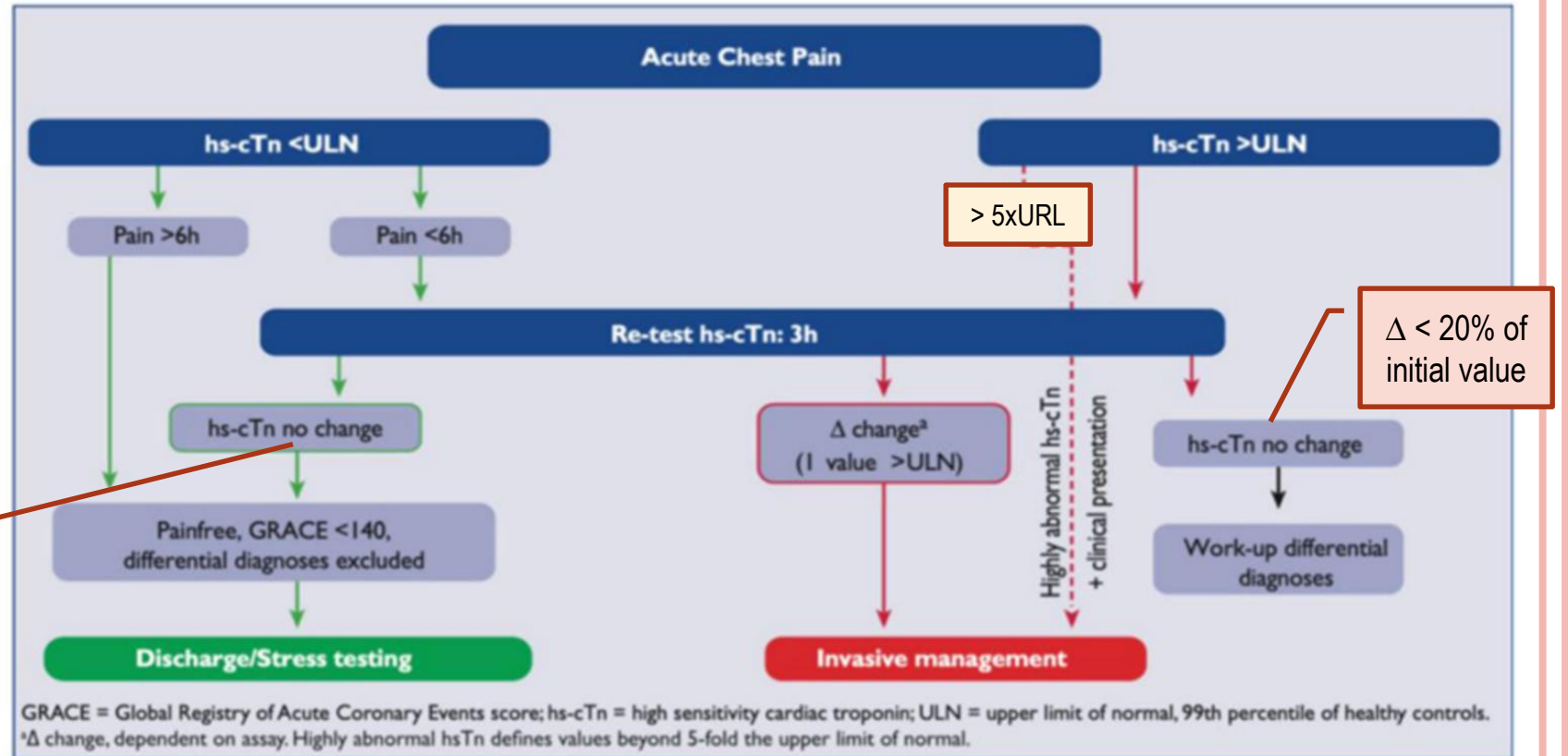


*Stable denotes $\leq 20\%$ variation of troponin values in the appropriate clinical context.

†Ischemia denotes signs and/or symptoms of clinical myocardial ischemia.

ESC 0/3-hour Algorithm

ESC guidelines year	Recommendations
2011	A rapid rule-out protocol (0 and 3 hours) is recommended when highly sensitive troponin tests are available. (Class IB)
2015	- A rapid rule-out protocol at 0 and 3 hours is recommended if high-sensitivity cardiac troponin tests are available. (Class IB) - A rapid rule-out and rule-in protocol at 0 and 1 hour is recommended if a high-sensitivity cardiac troponin test with a validated 0/1h algorithm is available. Additional testing after 3 to 6 hours is indicated if the first 2 troponin measurements are not conclusive and the clinical condition is still suggestive of ACS. (Class IB)

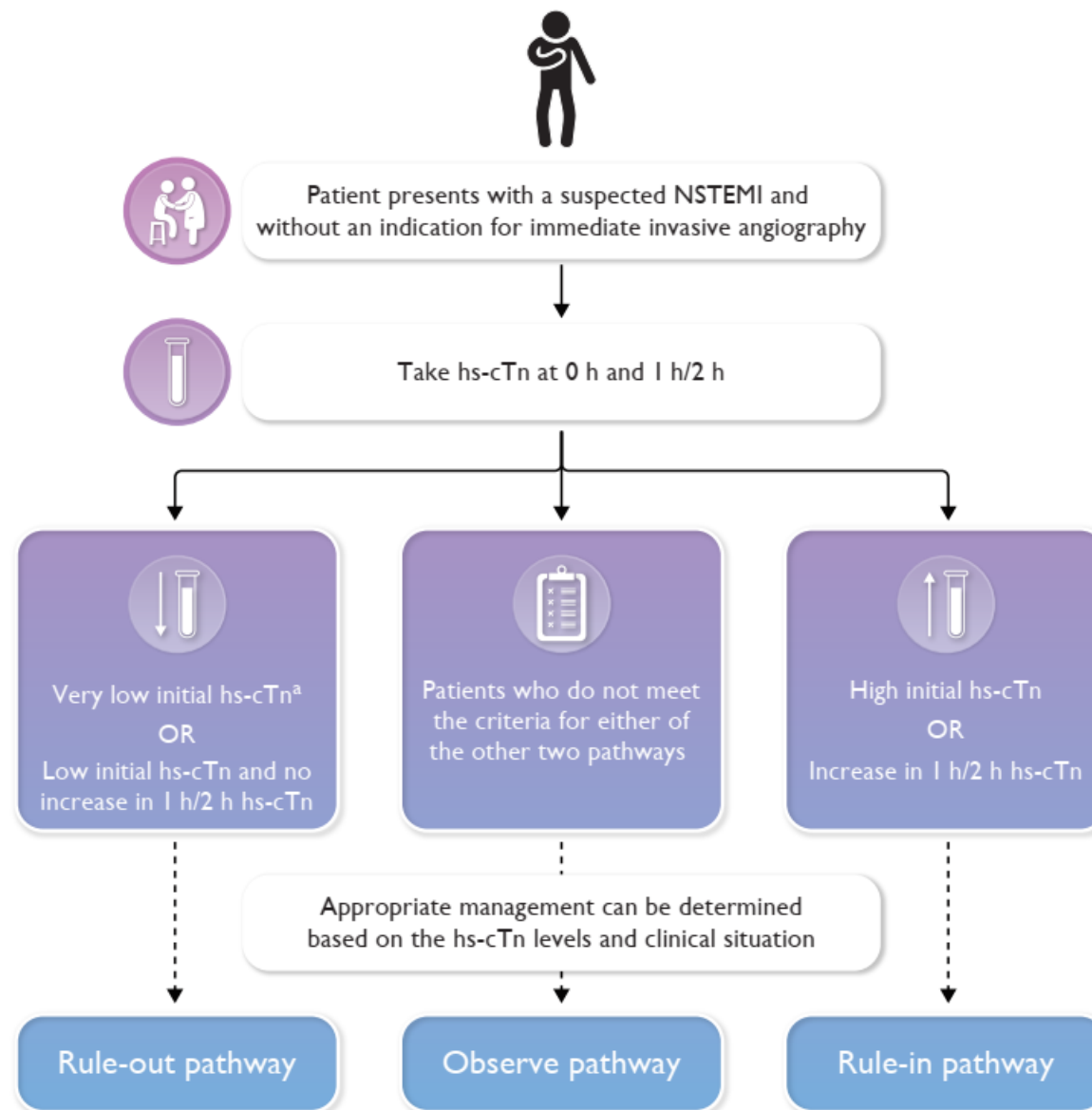
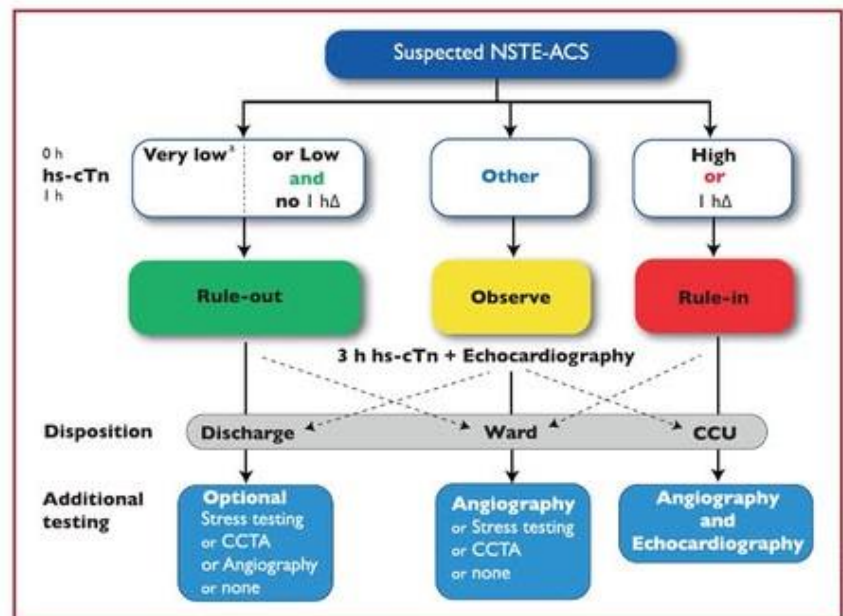


Δ < 50% of URL

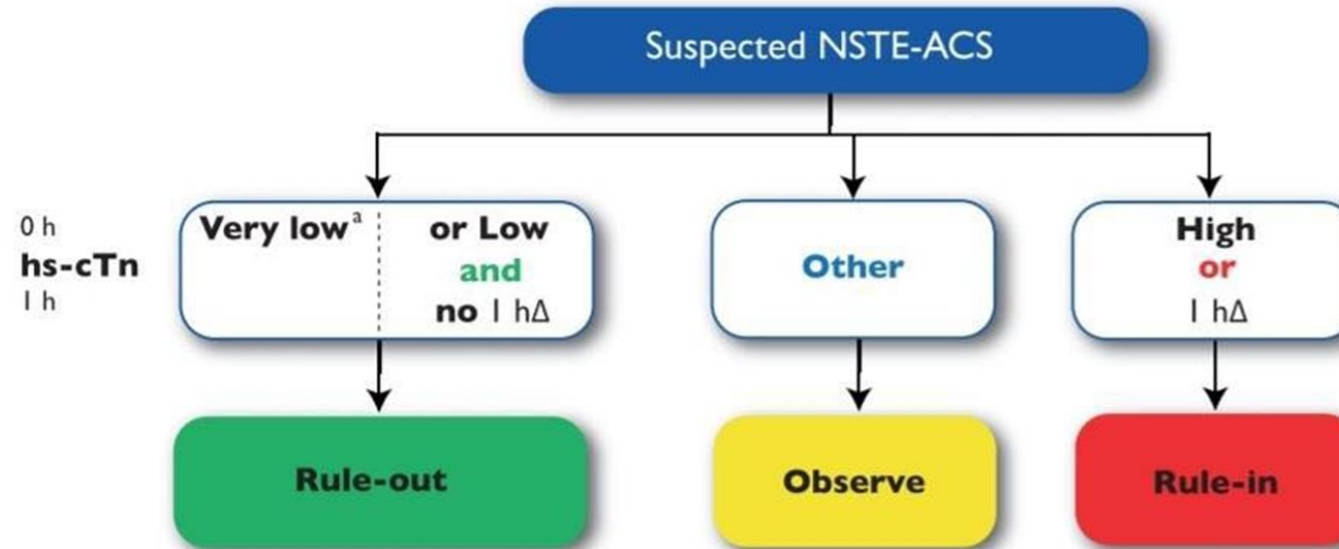
Δ < 20% of initial value

ESC guidelines year	Recommendations
2020	▪ The ESC 0/1h algorithm with blood sampling at 0 and 1 hour is recommended if a hs-cTn test with a validated 0/1h algorithm is available. (Class IB) ▪ Additional testing after 3 hours is recommended if the first 2 cardiac troponin measurements of the 0/1h algorithm are not conclusive and the clinical condition is still suggestive of ACS. (Class IB) ▪ As an alternative to the ESC 0/1h algorithm, it is recommended to use the ESC 0/2h algorithm with blood sampling at 0 and 2 hours, if a hs-cTn test with a validated 0/2h algorithm is available. (Class IB) ▪ As an alternative to the ESC 0/1h algorithm, a rapid rule-out and rule-in protocol with blood sampling at 0 and 3 hours should be considered, if a high-sensitivity (or sensitive) cardiac troponin test with a validated 0/3h algorithm is available. (Class IIa-B)

2023 ESC Guidelines for the Management of Acute Coronary Syndromes



ESC 0/1-hour Algorithm



Assay specific cut-off levels in ng/L within the 0/1 h algorithms

0 h/1 h algorithm	Very low	Low	No 1hΔ	High	1hΔ
hs-cTn T (Elecsys; Roche)	<5	<12	<3	≥52	≥5
hs-cTn I (Architect; Abbott)	<4	<5	<2	≥64	≥6
hs-cTn I (Centaur; Siemens)	<3	<6	<3	≥120	≥12
hs-cTn I (Access; Beckman Coulter)	<4	<5	<4	≥50	≥15
hs-cTn I (Clarity; Singulex)	<1	<2	<1	≥30	≥6
hs-cTn I (Vitros; Clinical Diagnostics)	<1	<2	<1	≥40	≥4
hs-cTn I (Pathfast; LSI Medience)	<3	<4	<3	≥90	≥20
hs-cTn I (TriageTrue; Quidel)	<4	<5	<3	≥60	≥8

ESC 0/2-hour Algorithm

Assay specific cut-off levels in ng/L within the 0/2 h algorithms

0 h/2 h algorithm	Very low	Low	No 2hΔ	High	2hΔ
hs-cTn T (Elecsys; Roche)	<5	<14	<4	≥52	≥10
hs-cTn I (Architect; Abbott)	<4	<6	<2	≥64	≥15
hs-cTn I (Centaur; Siemens)	<3	<8	<7	≥120	≥20
hs-cTn I (Access; Beckman Coulter)	<4	<5	<5	≥50	≥20
hs-cTn I (Clarity; Singulex)	<1	TBD	TBD	≥30	TBD
hs-cTn I (Vitros; Clinical Diagnostics)	<1	TBD	TBD	≥40	TBD
hs-cTn I (Pathfast; LSI Medience)	<3	TBD	TBD	≥90	TBD
hs-cTn I (TriageTrue; Quidel)	<4	TBD	TBD	≥60	TBD

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These cut-offs apply irrespective of age and renal function. Optimized cut-offs for patients above 75 years of age and patients with renal dysfunction have been evaluated, but not consistently shown to provide better balance between safety and efficacy as compared to these universal cut-offs.^{35,36,69} The algorithms for additional assays are in development.

hs-cTn = high-sensitivity cardiac troponin; TBD = to be determined.^{35–37,39,40,68,69,75–84}



Other Usefulness of cTn

Predictive/Prognosis marker in Mortality after Cardiac or Non-cardiac surgery

Prognostic relevance of troponin elevation after non-cardiac surgery

Taking a look at the VISION study

Vascular Events In Noncardiac Surgery Patients Cohort Evaluation (VISION)



Countries:
North and South America,
Africa, Asia, Australia, Europe

Study Objective:
Evaluate the relationship
between

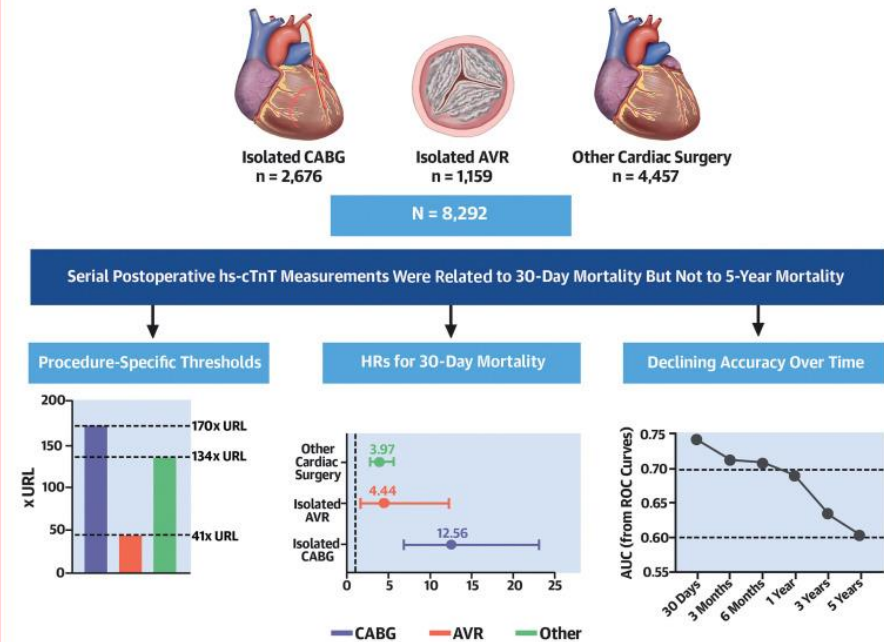
peroperative
troponin
measurements and
30-day
mortality in
patients

Association of High-Sensitivity Cardiac Troponin T With 30-Day and 5-Year Mortality After Cardiac Surgery



Leo Pölzl, MD,^a Clemens Engler, MD,^a Philipp Sterzinger, MSc,^b Ronja Lohmann,^a Felix Nägele, MD,^a

CENTRAL ILLUSTRATION: Defining Cardiac Troponin T Thresholds for Prediction of 30-Day Mortality



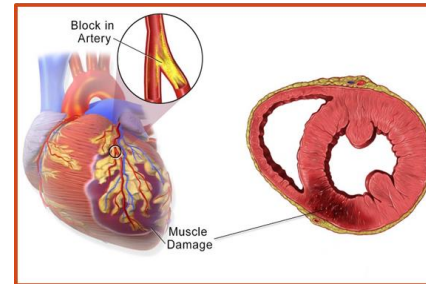
Pölzl L, et al. J Am Coll Cardiol. 2023;82(13):1301-1312.

Outline



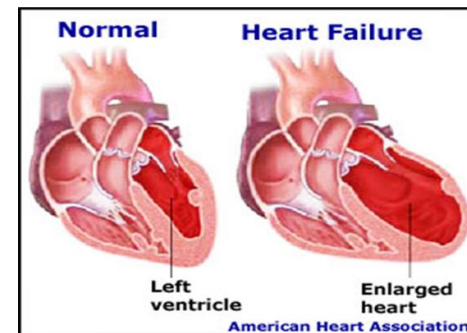
○ Biomarkers for Myocardial Necrosis

Acute Myocardial Infarction



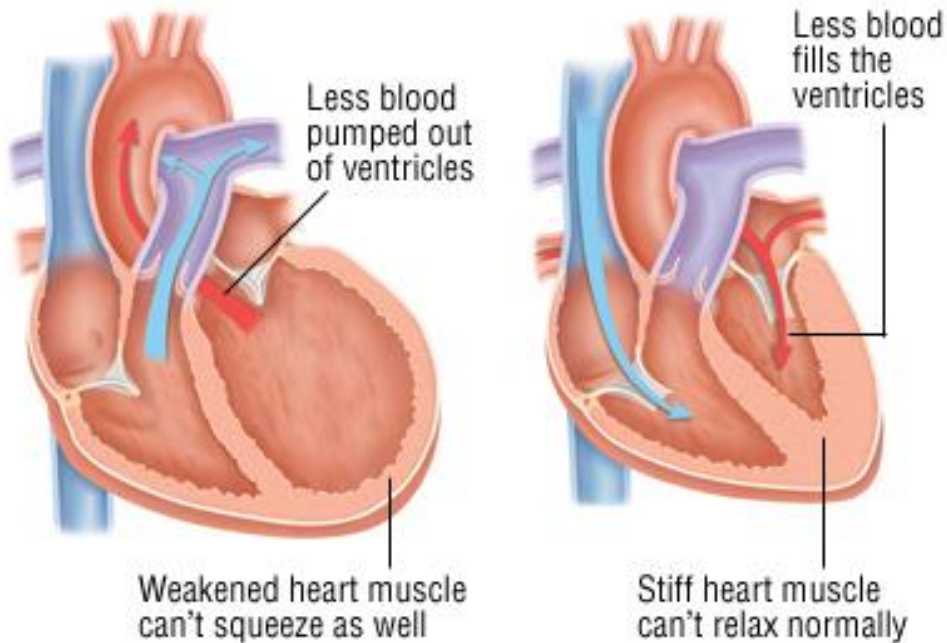
○ Biomarkers for *Myocardial Stretching* (Hemodynamic stress)

Heart Failure



Heart Failure (HF)

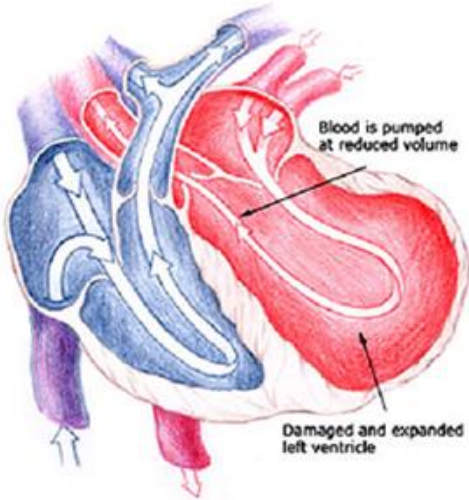
The joint American College of Cardiology: “A complex clinical syndrome that can result from any structural or functional cardiac disorder that impairs the ability of the ventricular to fill with or eject blood”



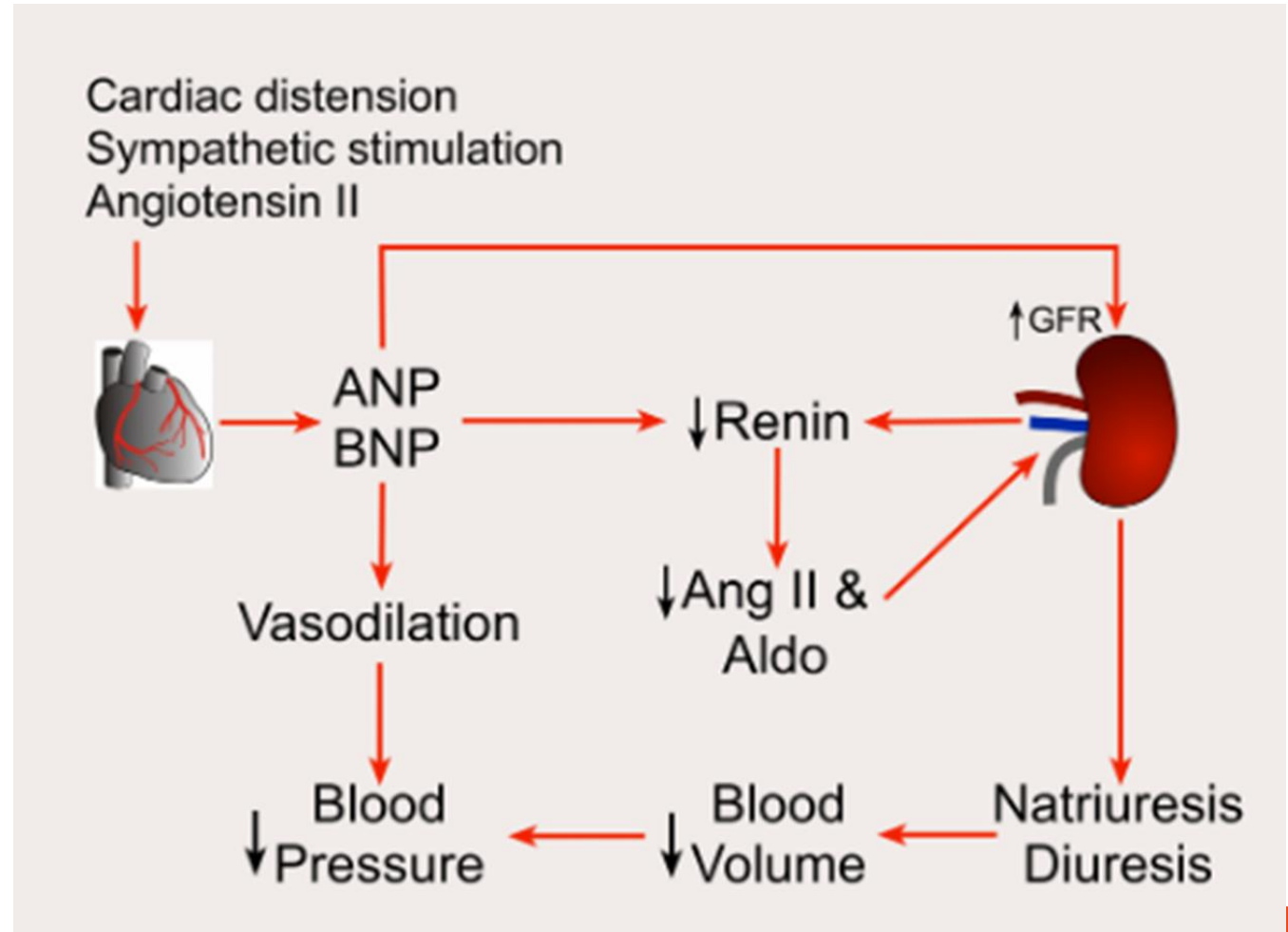
In some people with heart failure, the heart muscle becomes weaker. It cannot pump as well. In other people, the heart muscle becomes stiff. As a result, the heart cannot fill with enough blood between heartbeats.



Heart Failure



- ❑ The left ventricle is damaged and expanded while the blood is pumped out of the heart at a reduced volume.
- ❑ Cardiac ventricles secrete **natriuretic peptides** under conditions of volume expansion, hypertension and stretching of ventricular walls



Heart Failure

TABLE 1. Natriuretic Peptides.

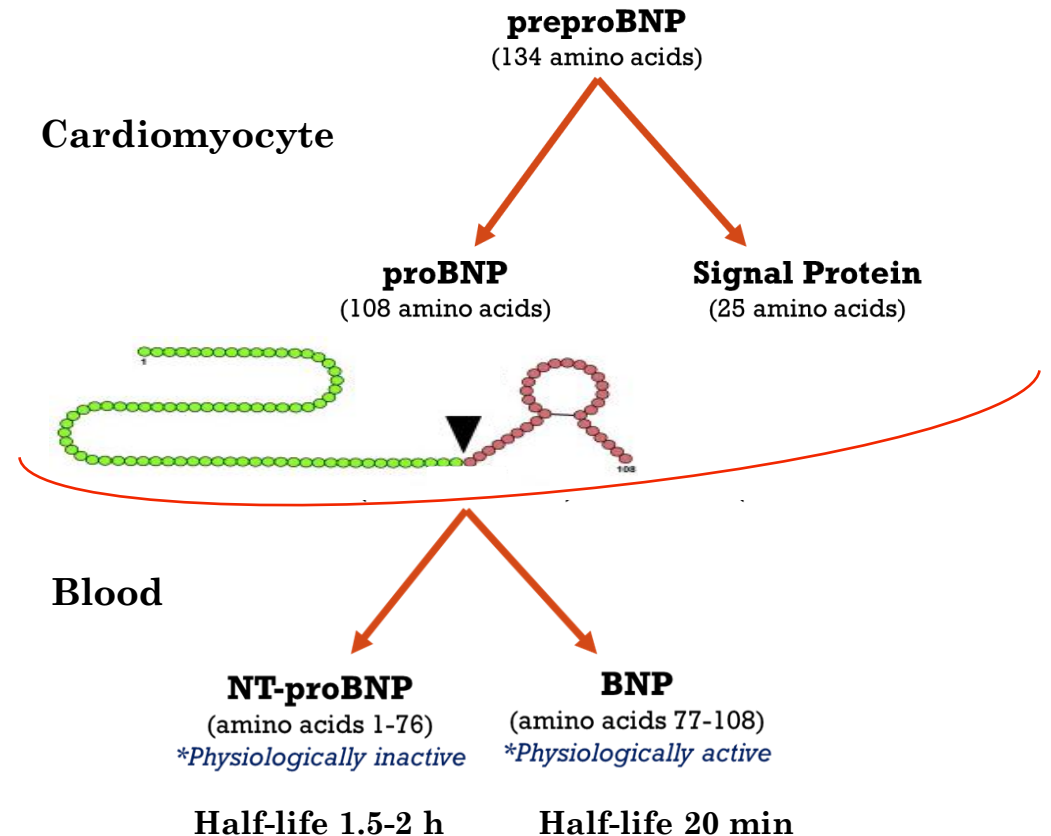
Peptide	Site of secretion	Biological activity	Half-life*
BNP	Ventricular myocardium	Vasodilation, natriuresis, diuresis; reduction of fibrosis and necrosis	20 min
ANP	Atrial myocardium	Blood pressure reduction; the inhibition of the renin-angiotensin-aldosterone system	2.5 min
CNP	Endothelial cells (the heart, brain, kidneys, vessels)	Vasodilation; inhibition of fibrosis, platelet aggregation, and tissue plasminogen activation	2.6 min

*The clinical significance of half-lives lies in their influence on diagnostic and therapeutic applications. BNP's longer half-life compared with ANP and CNP makes it particularly suitable for monitoring heart failure progression and treatment response; ANP, atrial natriuretic peptide; BNP, B-type natriuretic peptide; CNP, C-type natriuretic peptide.

Current Biomarkers for Hemodynamic Stress / Myocardial Stretching / Myocardial Wall Stress

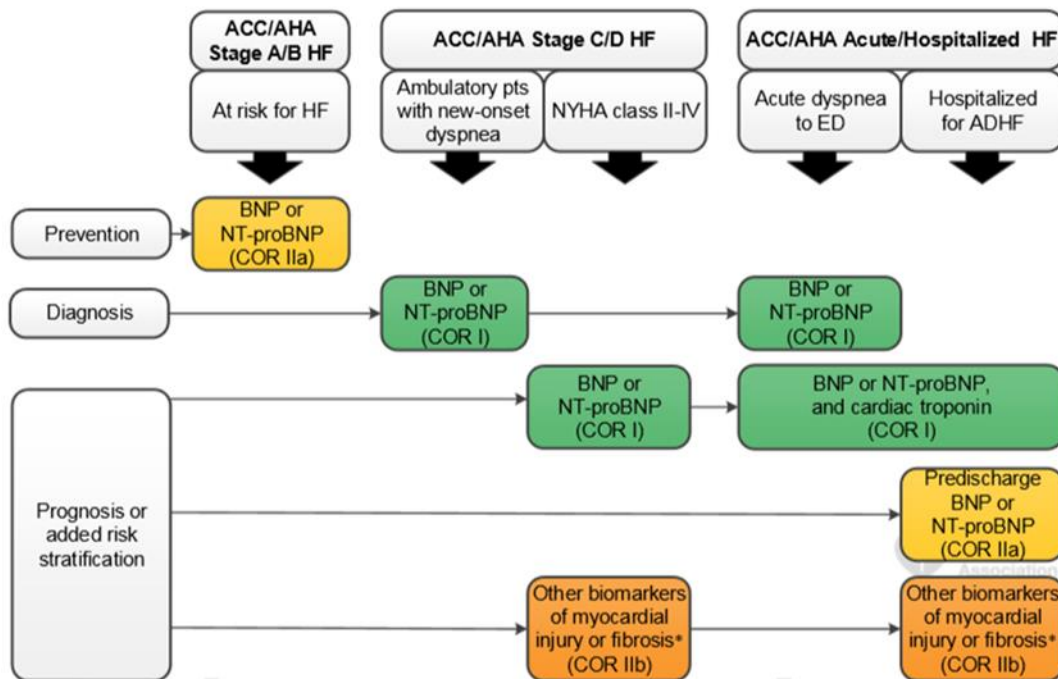
- ☐ B-type natriuretic peptide (BNP)
- ☐ N-terminal pro brain natriuretic peptide (NT-proBNP)

- BNP → **B-type** natriuretic peptide
- 32-amino acid polypeptide
- circulating & local hormone
- Ventricular myocyte



BNP & NT-proBNP in Heart Failure

- ❖ **BNP & NT-proBNP:** starting to rise within 2 h after HF
- ❖ **NT-proBNP:** remaining in blood 12-16 h after HF (longer half-life than BNP)
- ❖ **Stability in sample (25°C)** → BNP ~4 h vs **NT-proBNP >3 days**
- ❖ **Lab investigation:** Chemiluminescence microparticle immunoassay (CMIA)



Clinical Uses

- ❖ **Preventive values**
- ❖ **Diagnostic values**; Acute & Chronic HF
- ❖ **Prognostic values** → Determine HF worsened conditions e.g. death, readmission, cardiac events *etc.*
- ❖ **Guiding and Monitoring Heart Failure Treatment**

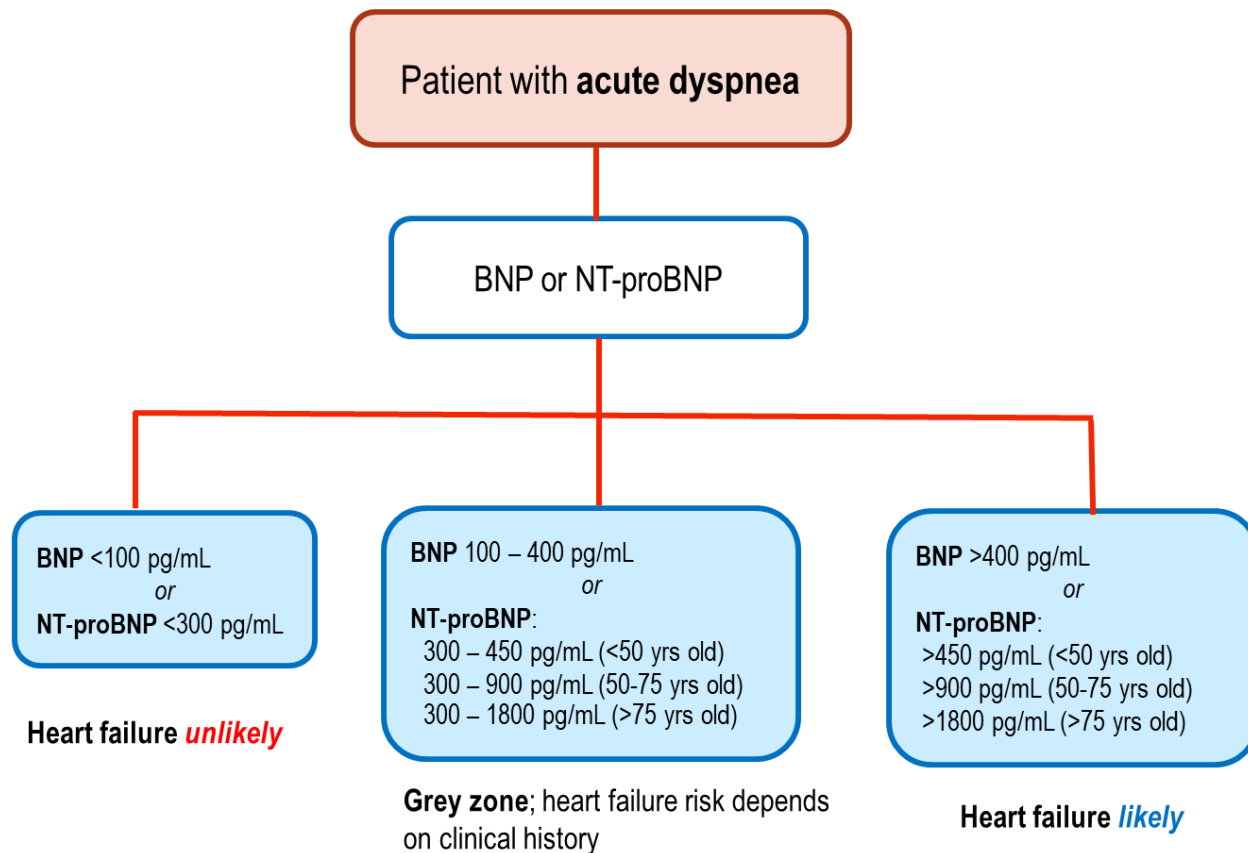


Table 1. A clinical scoring system for diagnosis of AHF

Predictor	Possible score	Your patient's score
Age > 75 y	1	
Orthopnea present	2	
Lack of cough	1	
Current loop diuretic use (before presentation)	1	
Rales on lung exam	1	
Lack of fever	2	
Elevated NT-proBNP*	4	
Interstitial edema on chest x-ray	2	
	14	Total =
Likelihood of heart failure	Low	0-5
	Intermediate	6-8
	High	9-14

AHF, acute heart failure; NT-proBNP, N-terminal pro brain natriuretic peptide.

*Elevated NT-proBNP was defined as > 450 pg/mL if age < 50 years and > 900 pg/mL if age > 50 years.¹¹

	Age, years	HF is unlikely	HF is possible but other diagnoses need to be considered	HF is very likely
Acute setting				
BNP	All	< 100 pg/mL	100-400 pg/mL	> 400 pg/mL
NT-proBNP	< 50	< 300 pg/mL	300-450 pg/mL	> 450 pg/mL
	50-75	< 300 pg/mL	450-900 pg/mL	> 900 pg/mL
	> 75	< 300 pg/mL	900-1800 pg/mL	> 1800 pg/mL
Ambulatory care setting				
BNP	All	< 50 pg/mL		
NT-proBNP	All	< 125 pg/mL		

BNP, B-type natriuretic peptide; HF, heart failure; NT-proBNP, N-terminal propeptide BNP.

Patient with **non-acute onset**

“Chronic heart failure”

BNP or NT-proBNP

BNP <35 pg/mL
or
NT-proBNP <125 pg/mL

Heart failure **unlikely**

BNP ≥35 pg/mL
or
NT-proBNP ≥125 pg/mL

Heart failure **likely**



Echocardiogram,
further work-up



BNP & NT-proBNP

Limitations

❑ Other physiological and pathophysiological factors affect to plasma concentrations of BNP and NT-proBNP.

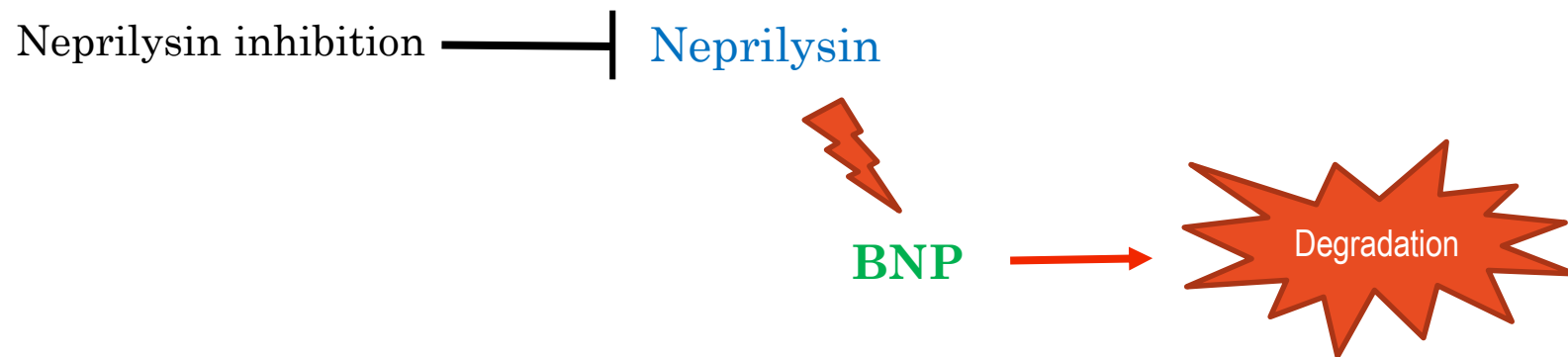
- **Age and Gender** ➤ Plasma BNP & NT-proBNP levels tend to be higher in women and older patients with or without cardiac dysfunction.
- **Obesity** ➤ Plasma BNP & NT-proBNP levels tend to be lower in obesity.
- **Renal dysfunction** ➤ Plasma BNP & NT-proBNP levels increase in chronic renal failure.



BNP & NT-proBNP

Limitations

- ❑ No increase of BNP and NT-proBNP in some types of heart failure
e.g., mitral stenosis, pericardium abnormal (cardiac tamponade)
- ❑ Drugs: angiotensin-neprilysin inhibitor (ARNI) → effect on the BNP level
→ No impact on the NT-proBNP level



Other use in AMI

NT-proBNP Cut-off Values for Risk Stratification in Acute MI and Comparison with Other Risk Assessment Scores

Shirley Siang Ning Tan ^{1,2}, Keng Tat Koh ³, Alan Yean Yip Fong ^{1,3}, Mohammad Adam bin Bujang ¹, Lee Len Tiong ¹, Yee Ling Cham ³, Kian Hui Ho ³, Chen Ting Tan ³, Chee Sin Khaw ³, Nor Hanim Mohd Amin ³, Yen Yee Oon ³, Asri Said ^{3,4}, Kent Ter Lau ⁵, Kar Ying Yong ⁵, Daniel Cheng Lee Pang ⁶, Chandan Deepak Bhavnani ^{3,6}, Ing Tien Wong ⁷, Francis Eng Pbeng Shu ^{3,8} and Tiong Kiam Ong ³

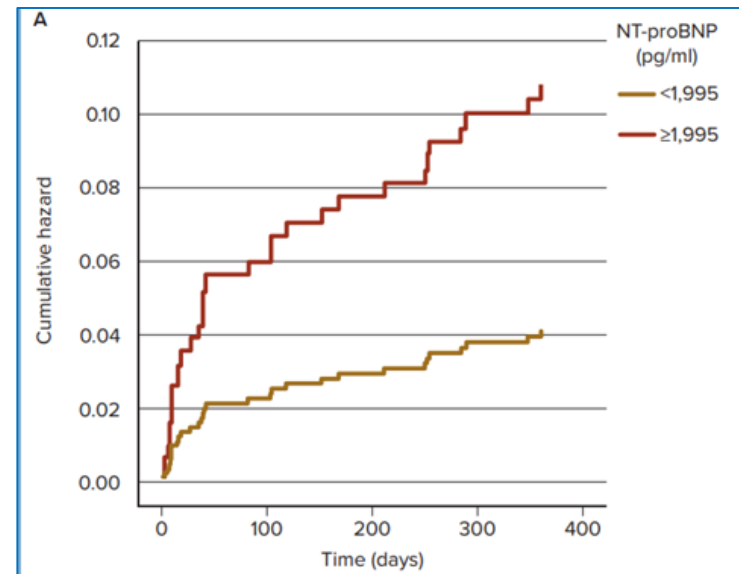
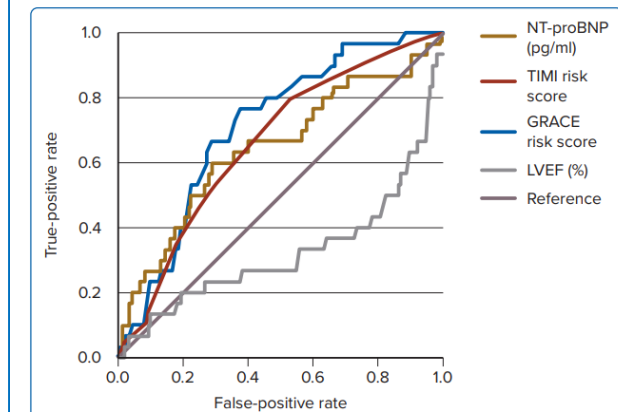


Figure 2: Comparing ROC Curve



GRACE = Global Registry of Acute Coronary Events; LVEF = left ventricular ejection fraction; NT-proBNP = N-terminal pro-brain natriuretic peptide; ROC = receiver operating characteristic; TIMI = Thrombolysis in MI.

Risk stratification

Measuring BNP or NT-proBNP plasma concentrations should be considered to gain prognostic information.

Read More

➤ เอกสารประกอบคำสอน

Laboratory Investigation in CVS: Cardiac Biomarkers

รศ.ดร.วิไลวรรณ ศรีวิมล
สาขาวิชาพยาธิวิทยา คณะแพทยศาสตร์
มหาวิทยาลัยสงขลานครินทร์

โรคหัวใจและหลอดเลือดเป็นสาเหตุการเสียชีวิตอันดับต้นๆ ของประชากรทั่วโลก โดย acute coronary syndrome (ACS) หรือกลุ่มภาวะกล้ามเนื้อหัวใจขาดเลือดเฉียบพลัน ซึ่งก็เป็นกลุ่มย่อยหนึ่งในโรคกล้ามเนื้อหัวใจขาดเลือด (ischemic heart disease; IHD) โดย ACS เกิดจากภาวะที่มีการอุดตันของเส้นเลือดแดง (coronary artery) ประกอบด้วยกลุ่มโรคต่างๆ คือ

1. Unstable angina (UA) คือ ภาวะกล้ามเนื้อหัวใจขาดเลือดแต่ยังไม่มีการตายของเซลล์กล้ามเนื้อหัวใจ มีอาการเจ็บแน่นหน้าอกที่รุนแรงเพิ่มขึ้น กินเวลานานขึ้น และอาการมักไม่บรรเทาเมื่อพัก
2. Myocardial infarction (MI) คือ ภาวะกล้ามเนื้อหัวใจตายจากการขาดเลือด แบ่งได้เป็น 2 ชนิด คือ ST elevation myocardial infarction (STEMI) และ non-ST elevation myocardial infarction (NSTEMI)

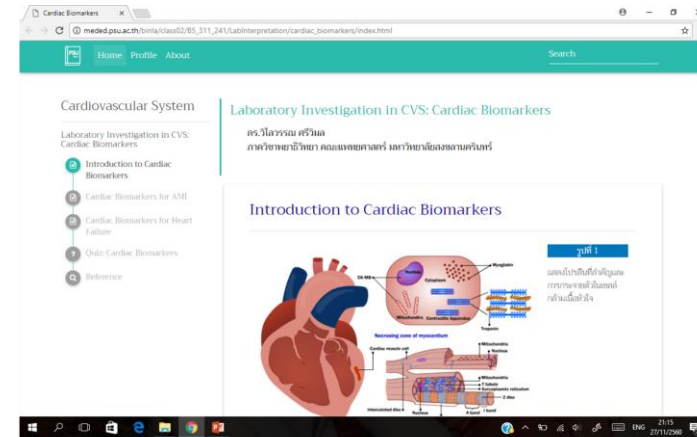
1. ภาวะกล้ามเนื้อหัวใจตายเฉียบพลัน (Acute Myocardial Infarction; AMI)

ภาวะกล้ามเนื้อหัวใจตายเฉียบพลัน หรือภาวะกล้ามเนื้อหัวใจตายจากการขาดเลือดแบบเฉียบพลัน (AMI) เป็นภาวะที่เซลล์กล้ามเนื้อหัวใจตายจากการขาดเลือดมาเลี้ยง ซึ่งมักเกิดจากหลอดเลือดที่นำเลี้ยงกล้ามเนื้อหัวใจเกิดการอุดตันกะทันหันจากก้อนลิ่มเลือด (thrombus) อาการมักจะเกิดอย่างเฉียบพลันโดยมีอาการเจ็บหน้าอก (chest pain) เป็นอาการหลัก กลุ่มผู้ป่วยที่มีกล้ามเนื้อหัวใจตายจากการขาดเลือด สามารถแบ่งออกเป็น 2 ลักษณะ คือ กล้ามเนื้อหัวใจตายชนิด NSTEMI และชนิด STEMI อัตราการเสียชีวิตของผู้ป่วยกลุ่มโรคนี้ค่อนข้างสูง ดังนั้นการวินิจฉัยโรคให้ได้อย่างถูกต้องรวดเร็วและรีบให้การรักษาอย่างเร่งด่วนจึงมีความสำคัญอย่างยิ่ง เพื่อช่วยในการรักษาได้ผลดียิ่งขึ้น ลดอัตราการเสียชีวิตและอัตราการเกิดโรคแทรกซ้อนอื่นๆ

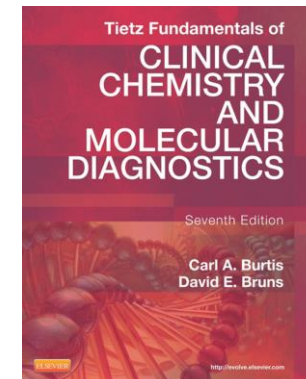
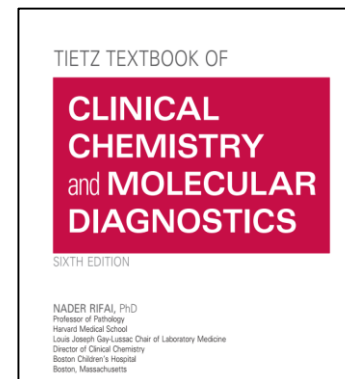
2. เกณฑ์การวินิจฉัย AMI

อาการเจ็บแน่นหน้าอก (chest pain) เป็นอาการหลักที่พบได้ใน AMI แต่อาการเจ็บแน่นหน้าอกสามารถเกิดจากสาเหตุอื่นๆ ที่ไม่ใช่ AMI ด้วยเช่นกัน เช่น โรคของหัวใจอื่นๆ ที่ไม่ใช่ AMI การบาดเจ็บของกล้ามเนื้อหน้าอก โรคของทางเดินหายใจ โรคปอด เป็นต้น การจะวินิจฉัย AMI จากอาการเจ็บแน่นหน้าอกจึงอาจนำไปสู่การวินิจฉัยผิดและการรักษาที่ไม่เหมาะสมได้ ทำให้ผู้ป่วยเสี่ยงต่อการได้รับผลแทรกซ้อน (adverse effect) ของยาสลายลิ่มเลือดที่ให้เพื่อการศึกษา AMI ได้ จึงต้องตรวจวินิจฉัยอย่างรอบคอบเพื่อให้เกิดประโยชน์ทางการรักษาและความปลอดภัยแก่ผู้ป่วย อย่างไรก็ดี ในผู้ป่วย AMI นั้น การรักษายาสลายลิ่ม

➤ Binla Book



<https://binlabook.medicine.psu.ac.th/pages/course/2>



Post-test Examination

เก็บคะแนนส่วน process → ทำ post-test exam ภายในเวลา **12.00** น ของวันนี้

- คะแนน ≥ 8 + ส่งภายในเวลาที่กำหนด → ได้คะแนนเต็ม
- คะแนน < 8 + ส่งภายในเวลาที่กำหนด → ได้คะแนน $\frac{3}{4}$ ของคะแนนเต็ม
- ส่งช้ากว่ากำหนด → ได้ $\frac{1}{2}$ ของคะแนนเต็ม
- ไม่ส่ง → ไม่มีคะแนน



Our Soul is for the Benefit of Mankind

